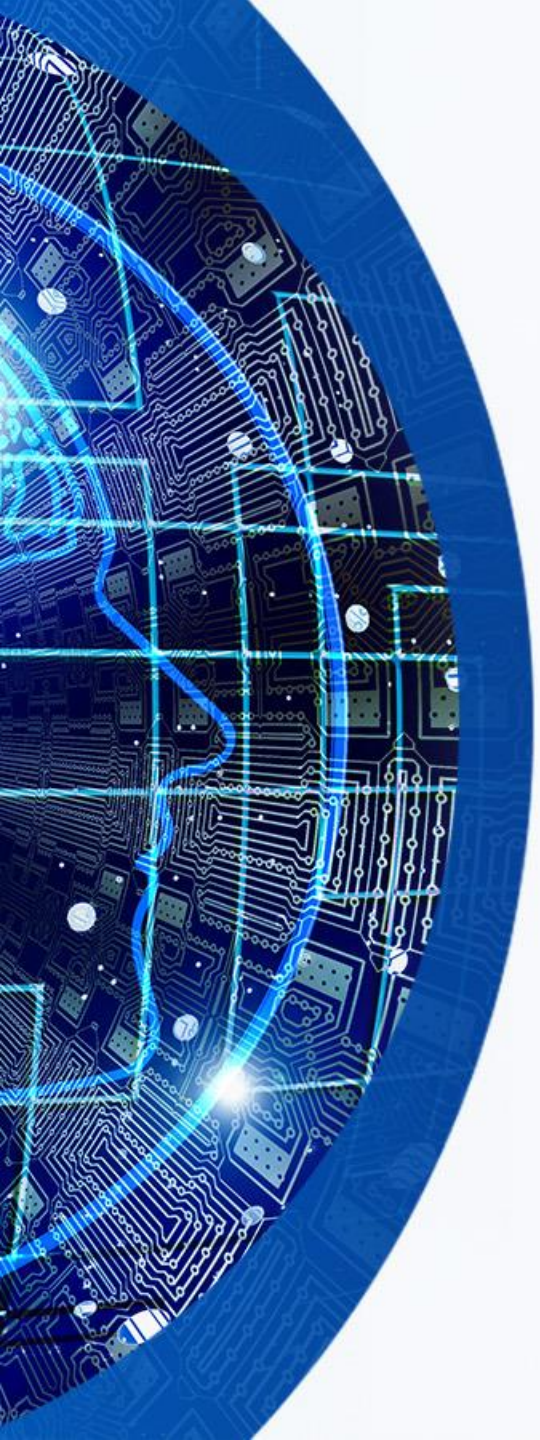
Natural Language Processing

Applications

NLP and Information retrieval (IR).

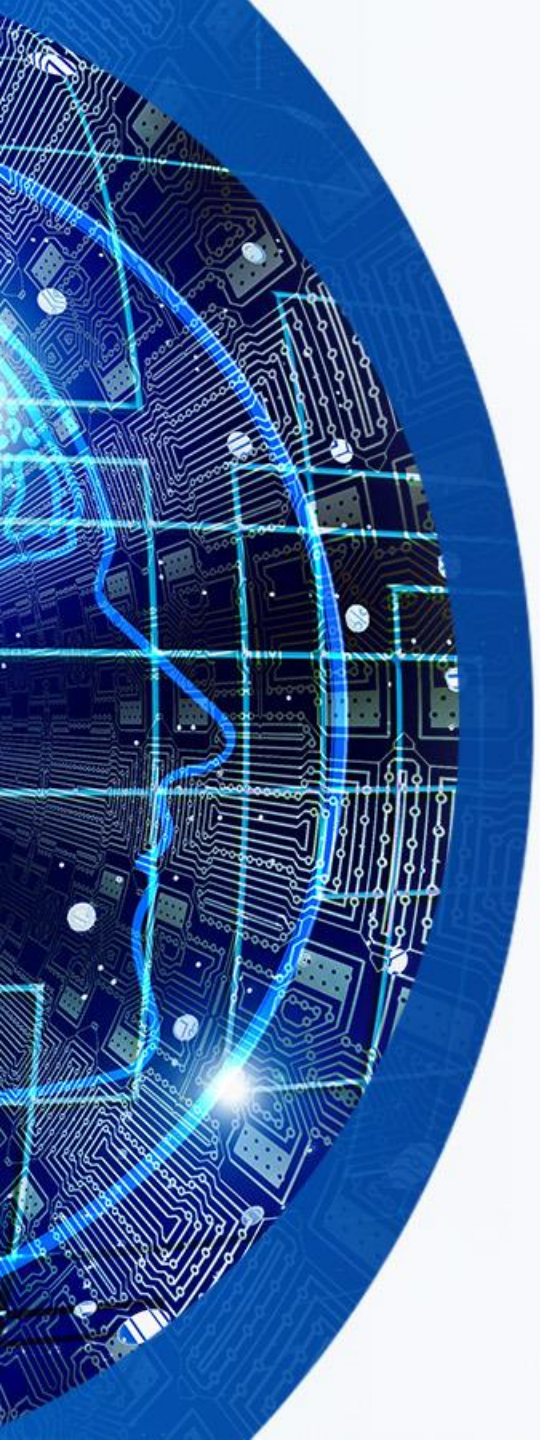
Outlines

- Intro to IR,
- IR & NLP
- Morphology,
- Orthographic Variation a
    - Syntax,
    - Semantics,



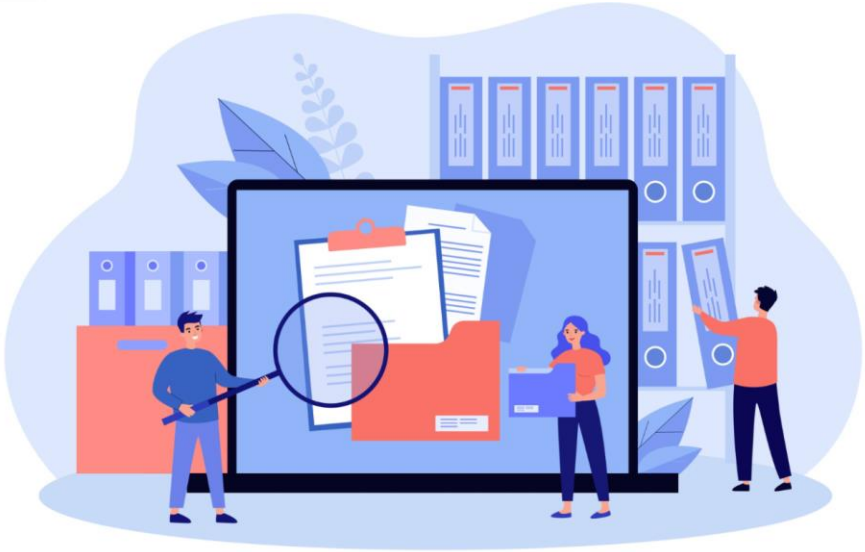
Intro to IR,

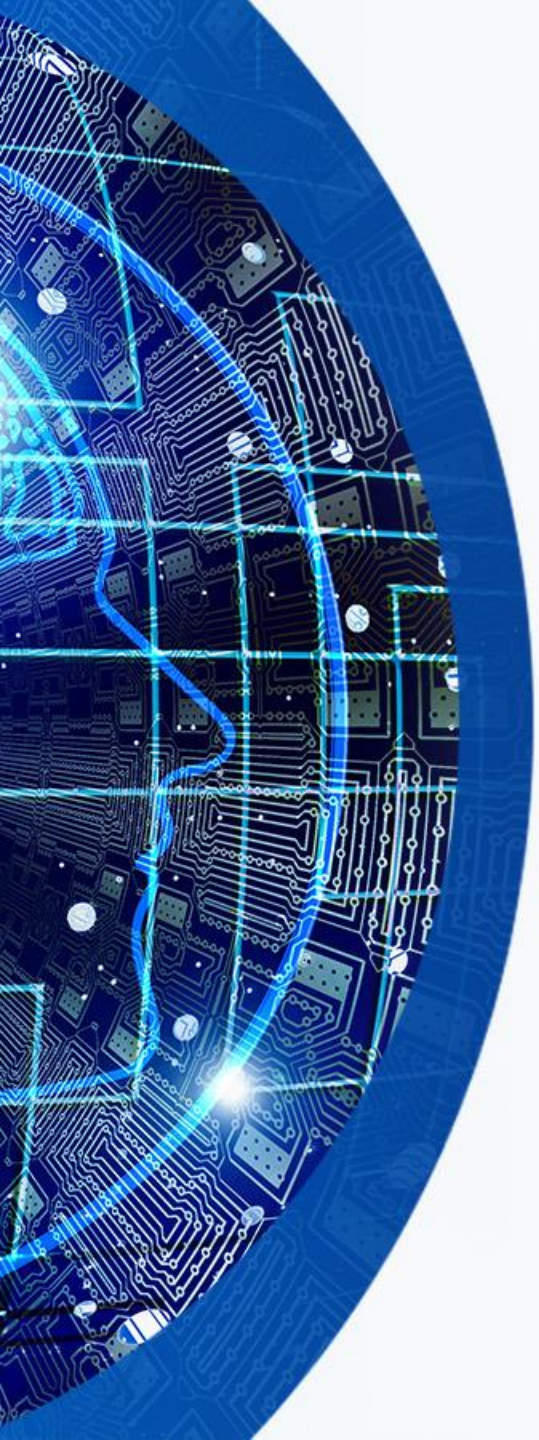
- The **Information Retrieval** (IR) domain can be viewed, to a certain extent, as a successful **applied domain** of NLP.
- The speed and scale of **Web take-up** around the world have been made possible by freely available and **effective search engines.**
- These tools are used by around **85% of Web** surfers when looking for some specific information.



Intro to IR,

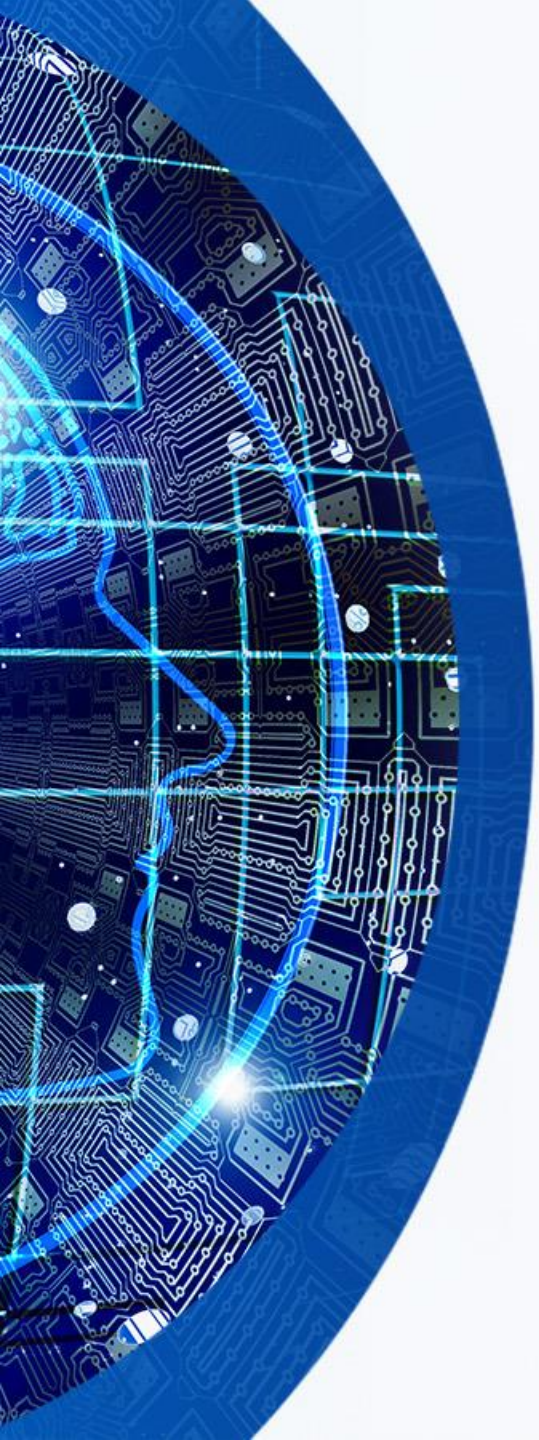
- IR deals with the **representation**, **storage**, **organization** of, and access to information items.
- These information items could be **references** to:
- real documents,
- documents themselves,
- or even single paragraphs,
- as well asWeb pages,
- spoken documents,
- images, pictures, music, video,
- etc





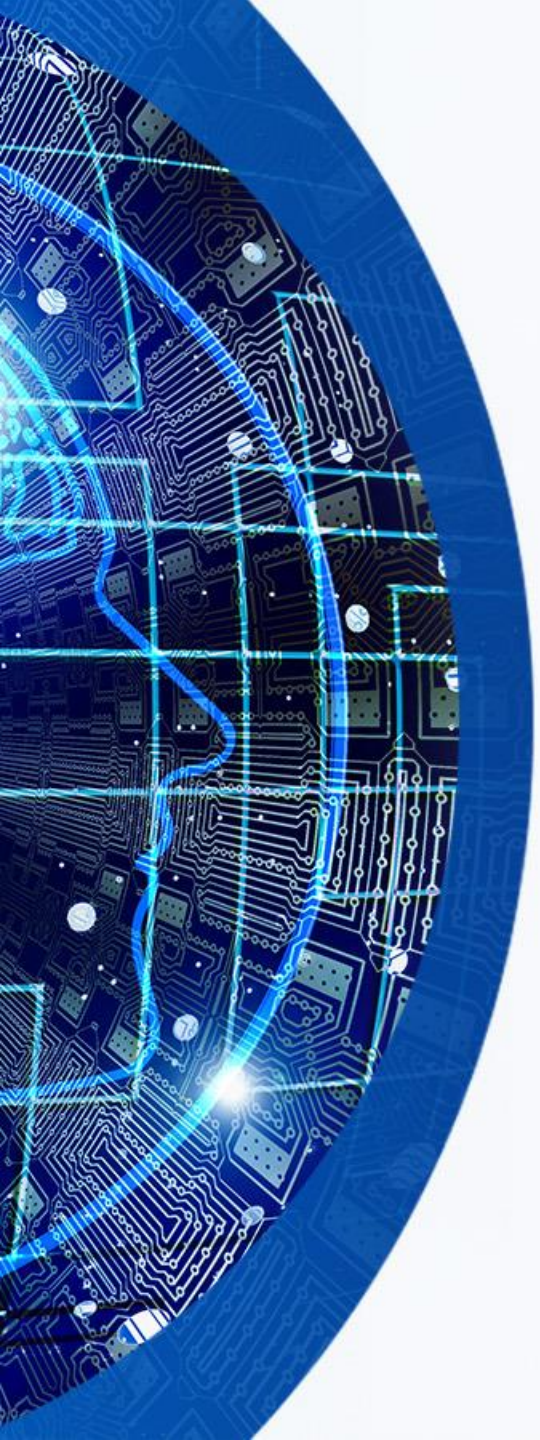
Intro to IR,

- One of the first and most important characteristics of IR is the fact that an IR system has to deal with **imprecise** and **incomplete descriptions** of both **user needs** and **documents** queried:
- in most cases, it is **impossible** to compute a precise and unambiguous representation for queries and documents.
- If SQL queries **cannot be ambiguous**, users sending queries to Web search engines tend to write **very short** and **ambiguous** descriptions of their **information needs**



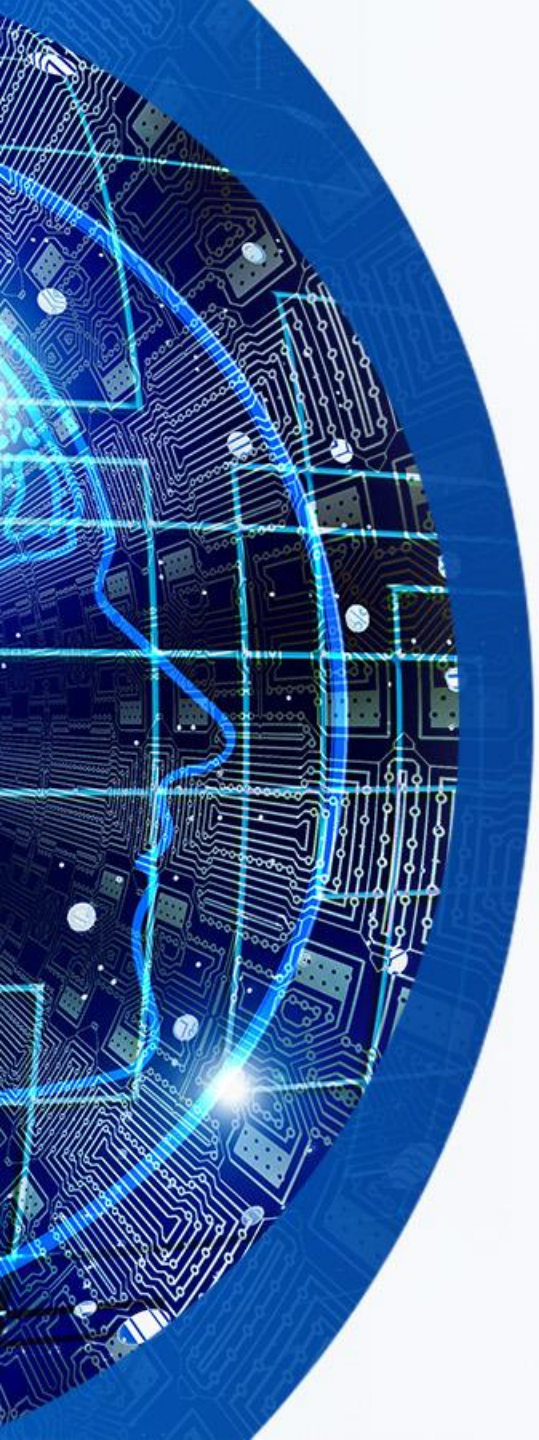
Intro to IR,

- Clearly, users do not **describe** their information needs with all the needed **details**, preferring the system to **suggest** some answers, and selecting, from these answers, the most appropriate items.
- The search process should be viewed more as a "**trial-and-error**" problem-solving approach than a direct "**query-response**" paradigm.
- Finally, and **contrary** to SQL-based search, the matching between the query and the information item is not **deterministic**: the system provides the **best possible answers** (best matches) by estimating their underlying **probability** of **relevance** to the submitted query.



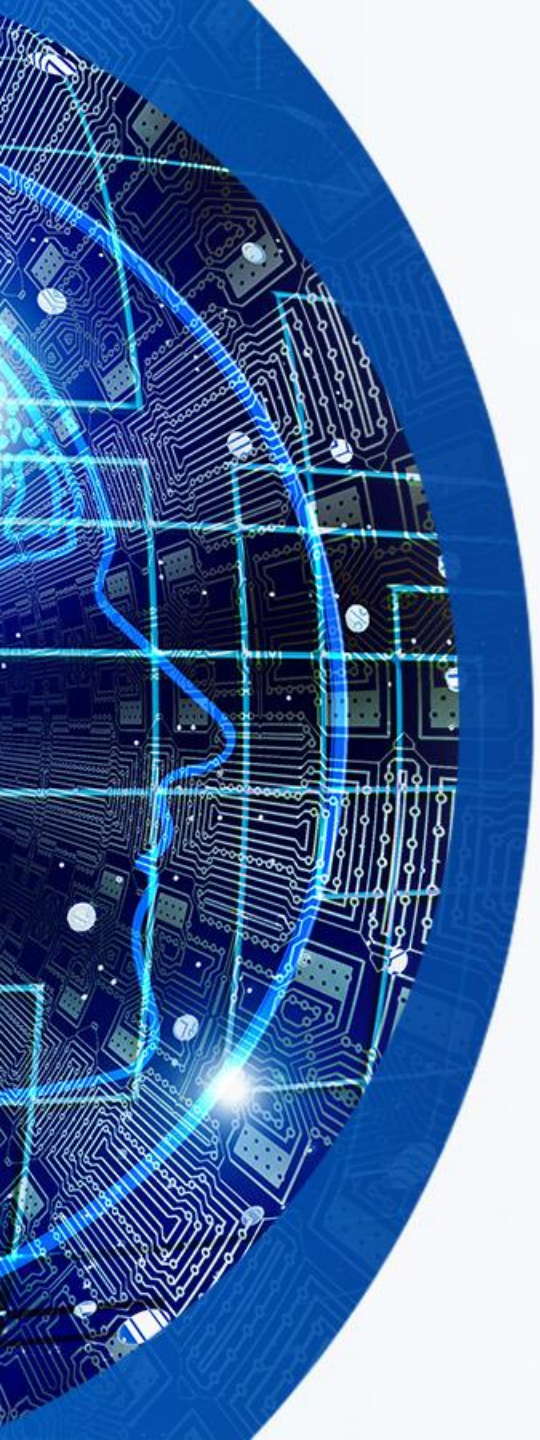
Intro to IR,

- The underlying difficulty in **retrieving** documents **relevant to a query** resides in the three aspects of all natural languages, namely, **polysemy, synonymy**, and, to a lesser extent, **spelling errors** and variations.
- In all natural languages, a given word may have more than **one precise meaning** (polysemy).
- Some of these senses could be **related**, but in other cases, the underlying meaning could **vary greatly.**
- For example, the word “bank” could be used to designate a financial institution, its building, a synonym of “**rely upon**” in the expression “**I’m your friend, you can bank o me,**” or the borders of a river.
- The last sense is of course not related to the previous three.



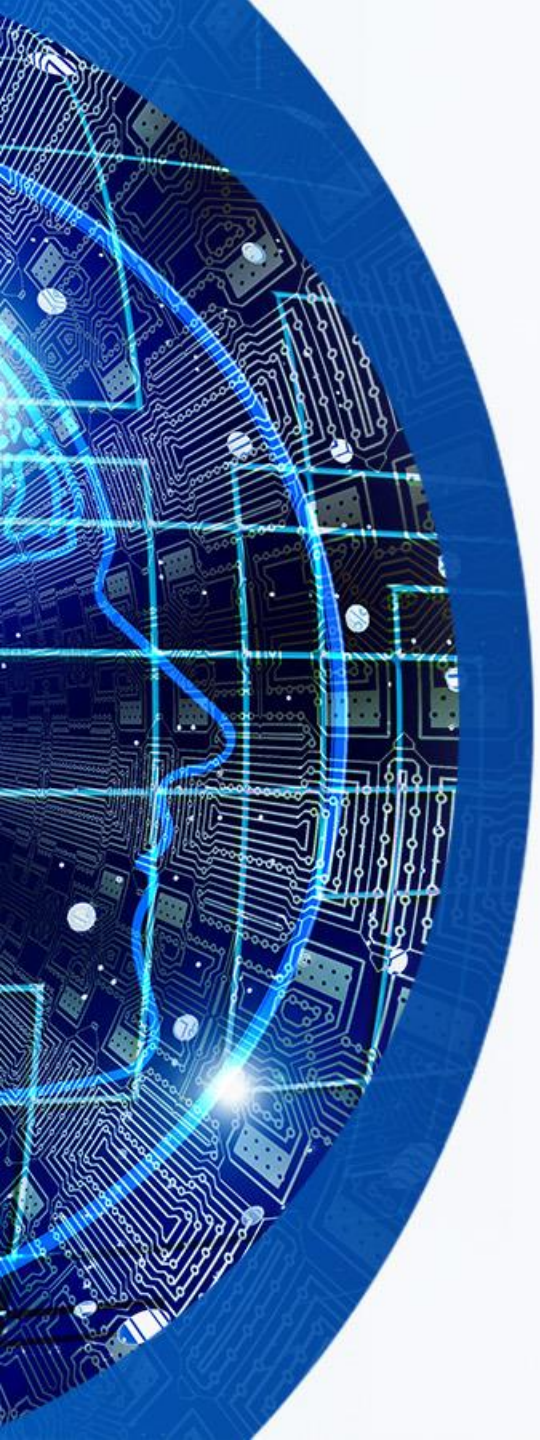
Intro to IR,

- With the word “Java,” the possible meanings are even less related (an island, coffee, a dance, a domestic fowl, a computer programming language).
- **Acronyms** are also subject of such ambiguity problems as, for example, BSE
  - Bovine Spongiform Encephalopathy,
  - Bombay Stock Exchange (or Boston, Beirut, Bahrain),
  - Breast Self-Examination,
  - Bachelor of Science in Engineering,
  - Basic Service Element, etc….



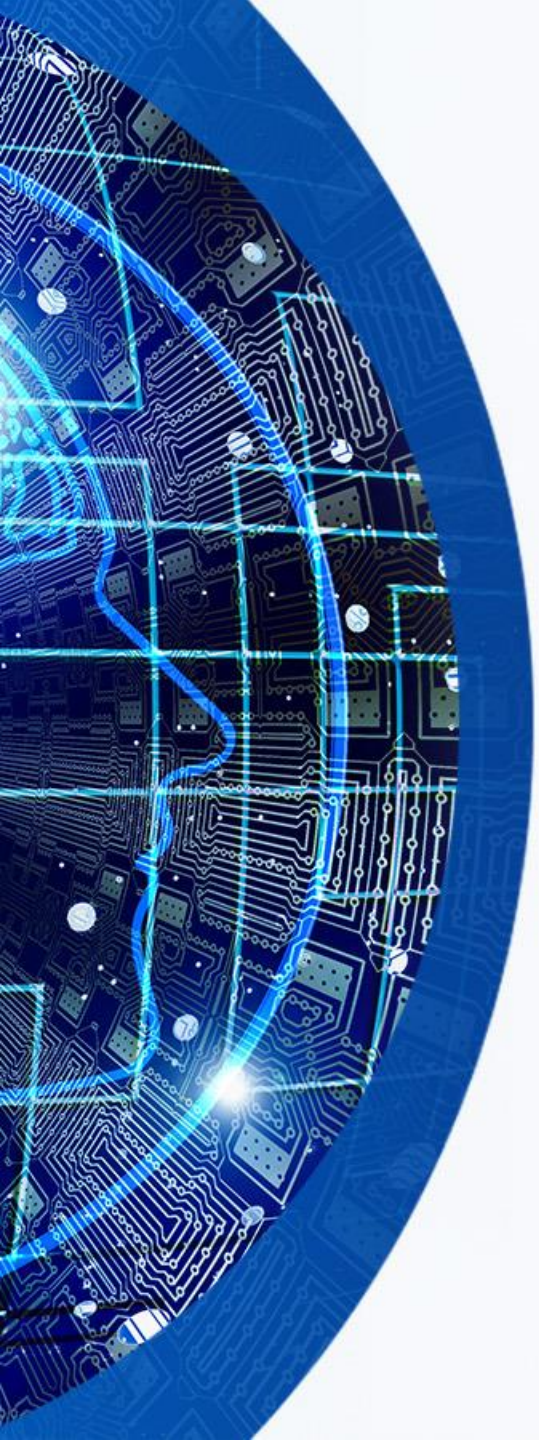
Intro to IR,

- Polysemy corresponds to one face of a coin.
- Synonymy, i.e., the fact that different words or expressions can be used to refer to the same object, is the second.
- For example, to refer to a given car **accident**, we can use the following terms:
  - “accident,” “event,” “incident,” “situation,” “problem,” “difficulty,” “unfortunate situation,” “the subject of your last letter,” “what happened last month,” ....



Intro to IR,

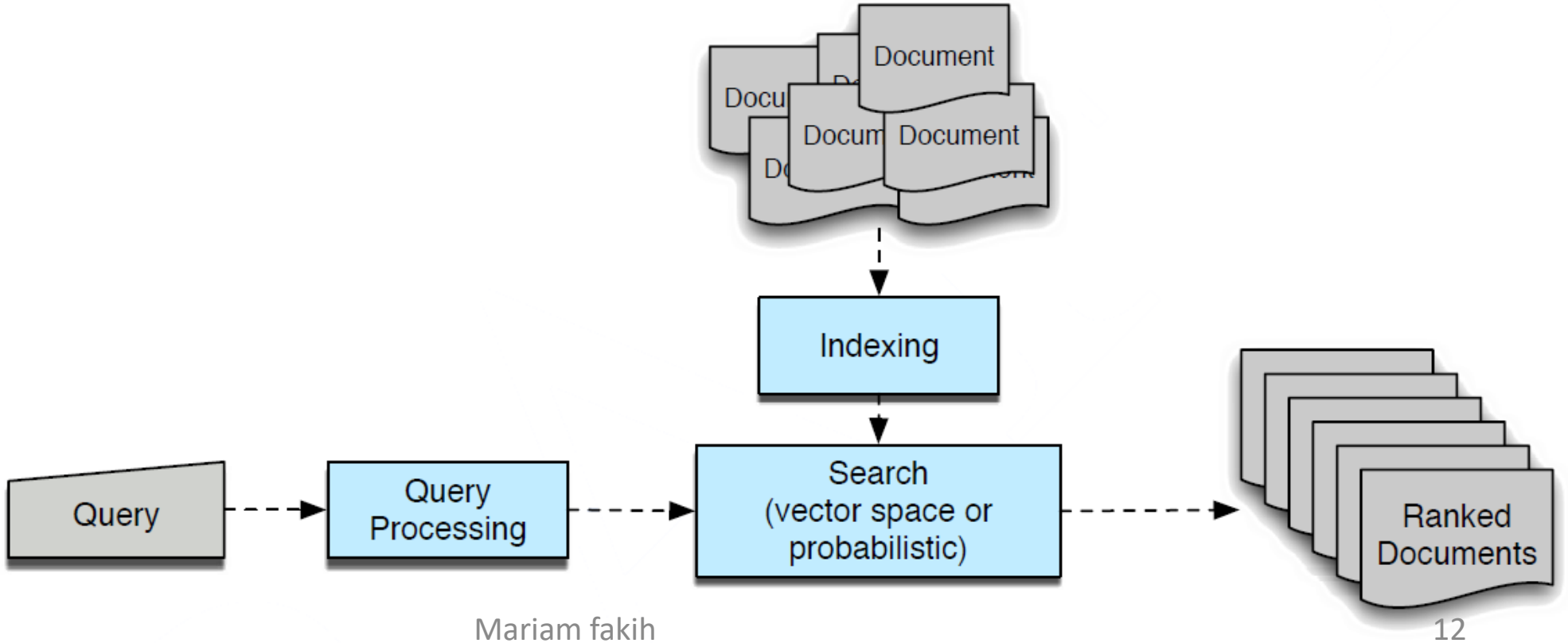
- In the example below, the expressions written in italics refer to the same person. It is also interesting to note that the string "Major" does not always refer to the same person:
- "*Mr Major* arrived in France today. *The Prime Minister* will meet the President tomorrow. *The Conservative leader* will then travel to Moscow where he will meet Mr Gorbachev. Mrs Major will join *her husband* in Russia, where *this son of a circus artist* is a relatively unknown figure."
  - As a result of these two main problems, the probability that two people use the same term to describe the same object is below 0.20.



Intro to IR,

- All search systems assume that similar words tend to correspond to similar meanings.
- Thus, when a query shares many terms with a document, this document is seen as an appropriate answer to the corresponding query.
- Search systems are thus based on a simple approach: **extract words** from documents and queries, **compare** the two sets of words, and **rank** the document according to this comparison.
- as the collection size increases, the achieved performance of such a process tends to increase.

Indexing


Document
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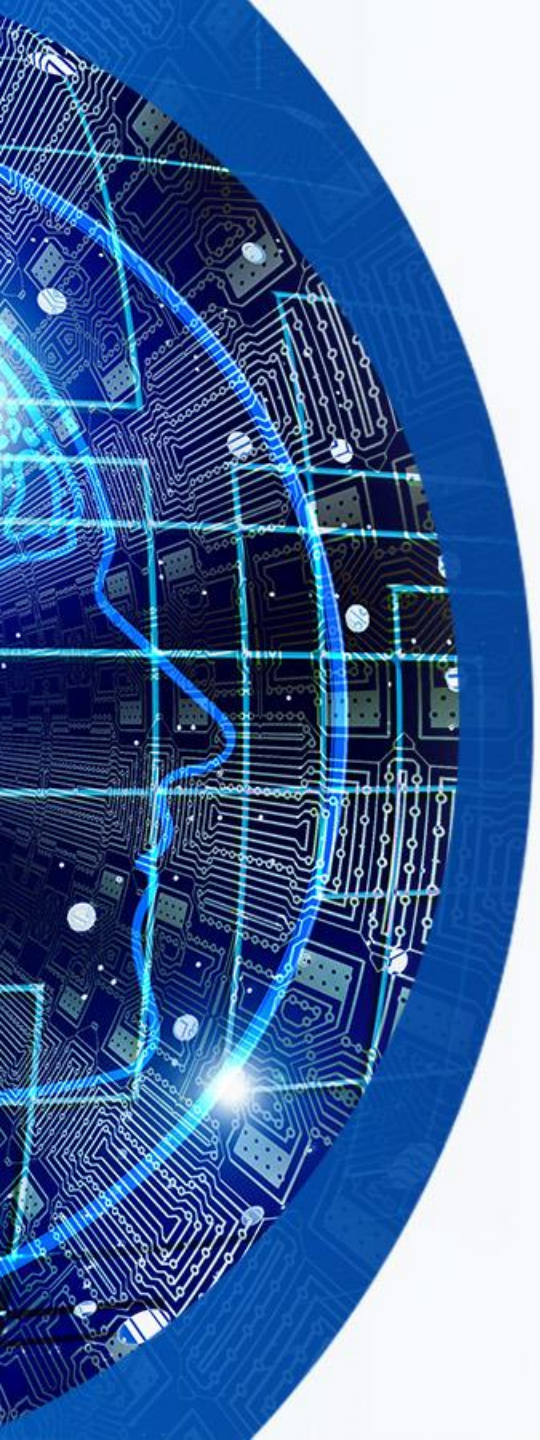
Docum
Document
Indexing
Query
Query Processing
Search (vector space or probabilistic)
Ranked Documents



Indexing

- the indexing process used to build documents or **query representations.**
- Effective search systems do not work **directly** with the documents (or the queries).
- They use different techniques and strategies to represent the main semantic aspects of the documents and queries.
- This process is called **indexing**.

Indexing

- Structure analysis and tokenization
- StopWord removal
- Morphological normalization
- Weighting

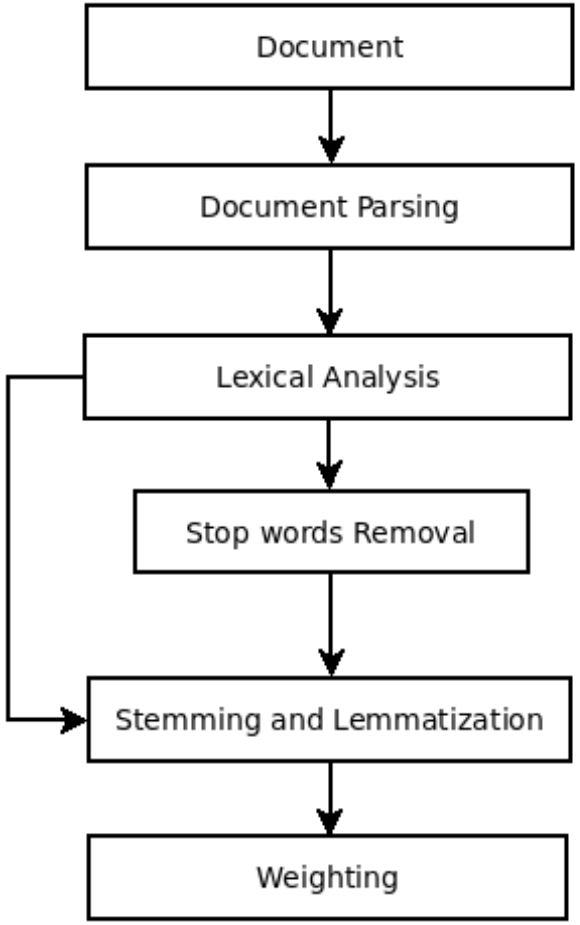
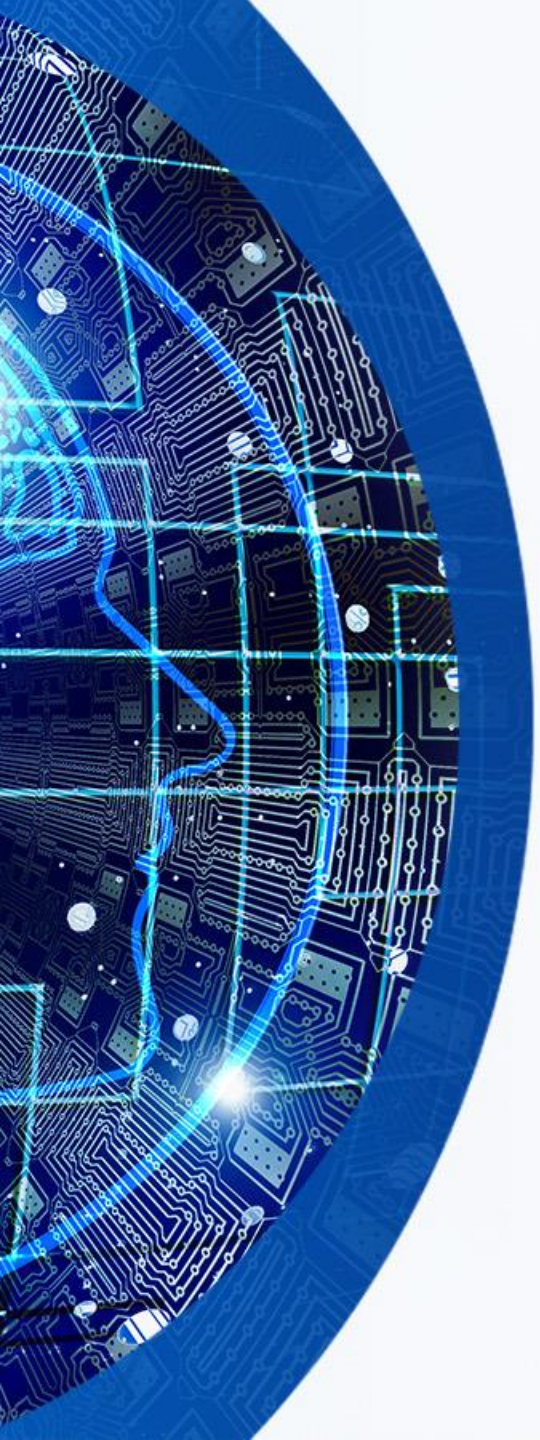


Fig. Processing of Textual data



Indexing

- **Structure analysis and tokenization**

  In this first step, documents are **parsed** so as to recognize their structure (title, abstract, section, paragraphs).

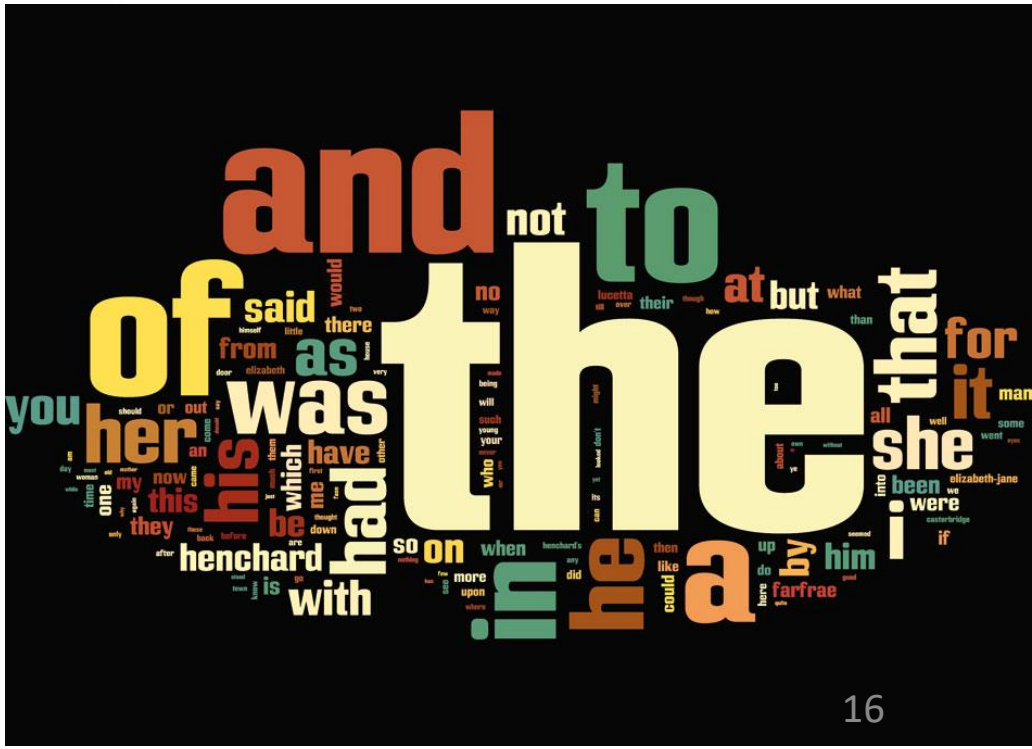
  For each relevant logical structure, the system then segments sentences into word tokens (hence the term **tokenization**).

Indexing

- **Stopword removal**

  In a second step, very frequent word forms such as
  - determiners “the”,
  - prepositions “from”,
  - conjunctions “and”,
  - pronouns “you”
  - and some verbal forms “is”,
  - etc.

  appearing in a **stopword** list are usually removed.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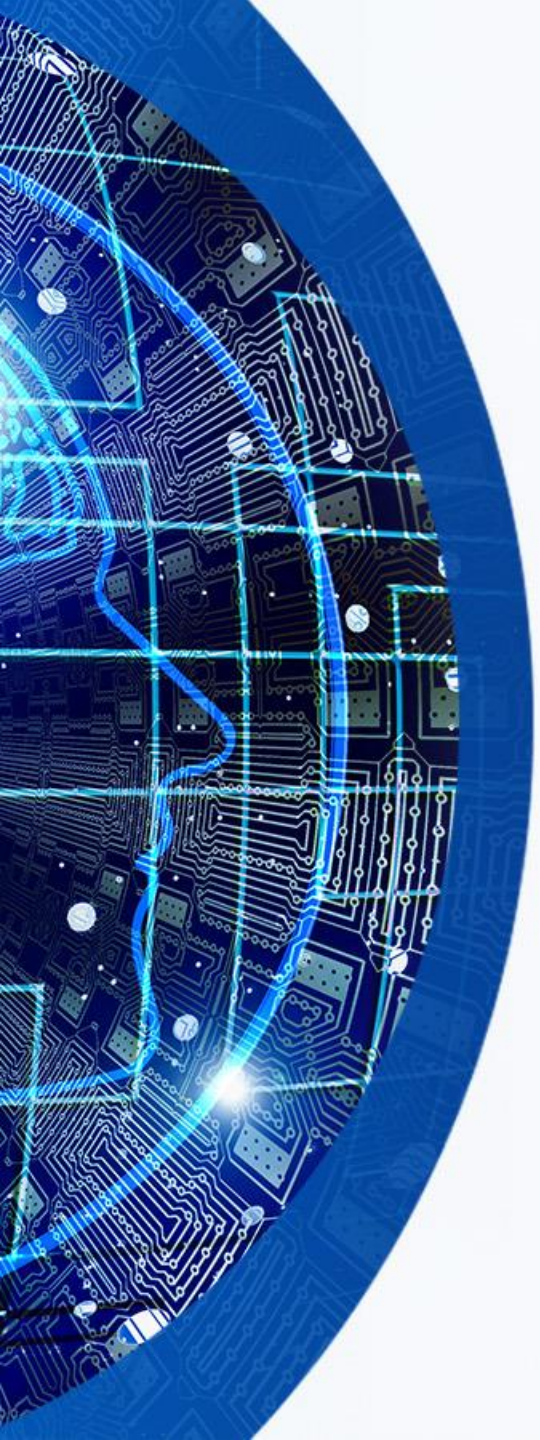
Indexing

•**Morphological normalization**

As a third step, an indexing procedure uses some type of morphological normalization in an attempt to conflate word variants into the same stem or root.

- **Stemming** procedures, which aim to identify the stem of a word and use it in lieu of the word itself, are by far the most common morphological normalization procedures used in IR.
- Grouping words having the same **root** under the same stem (or indexing term)may **increase the success rate** when matching documents to a query.



Indexing

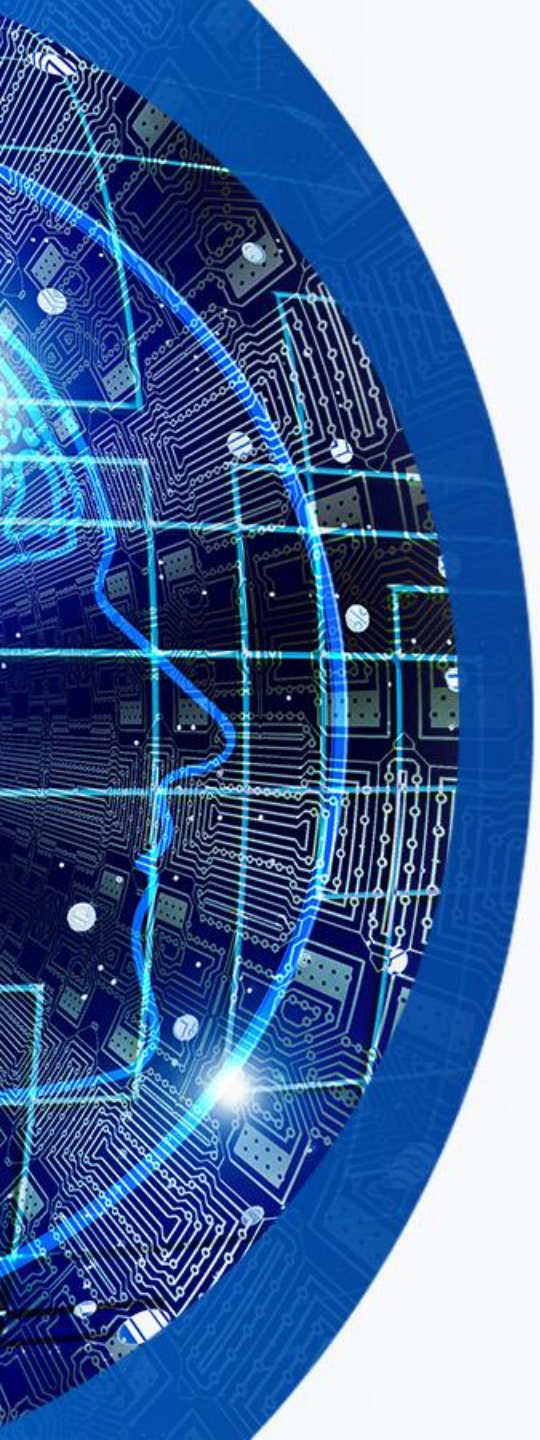
- **Weighting**

➢ After all last steps applied, from the sentence "*In 1969, the IBM-360 computer was one of the first third generation computers*," we can obtain the following indexing units:

➢ **"IBM-360," "first," "third," "generat," "comput."**

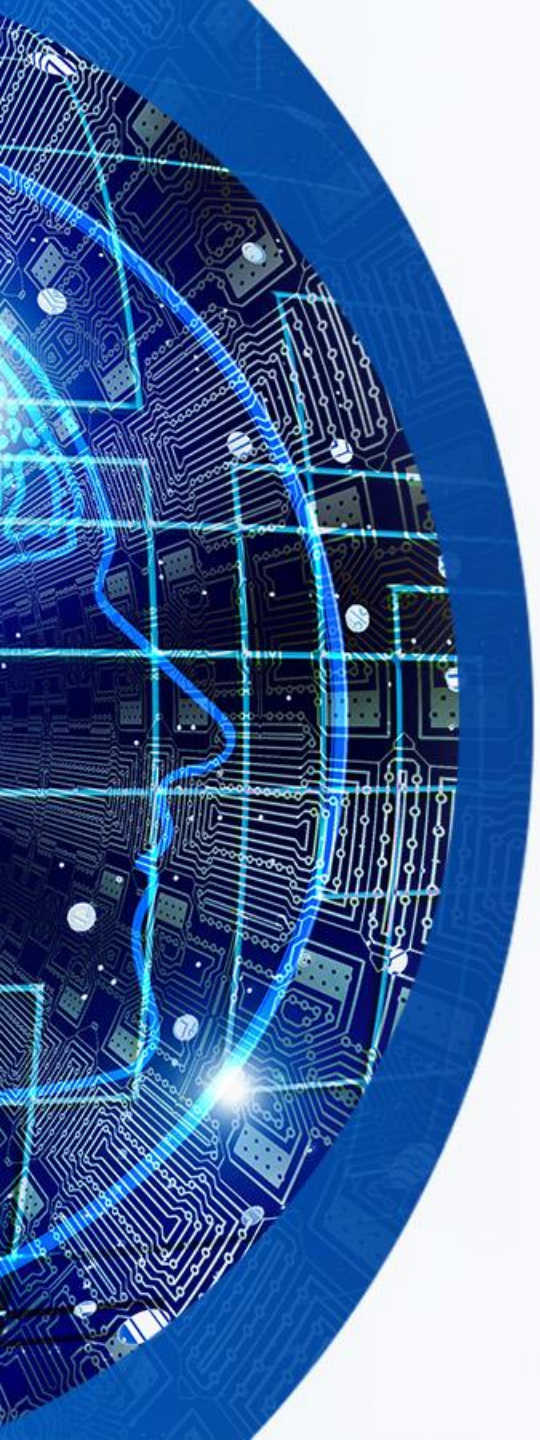
➢ This result corresponds to a **binary indexing** scheme within which each document is **represented** by a set of (stemmed) keywords

➢ without any **weight** assigned.



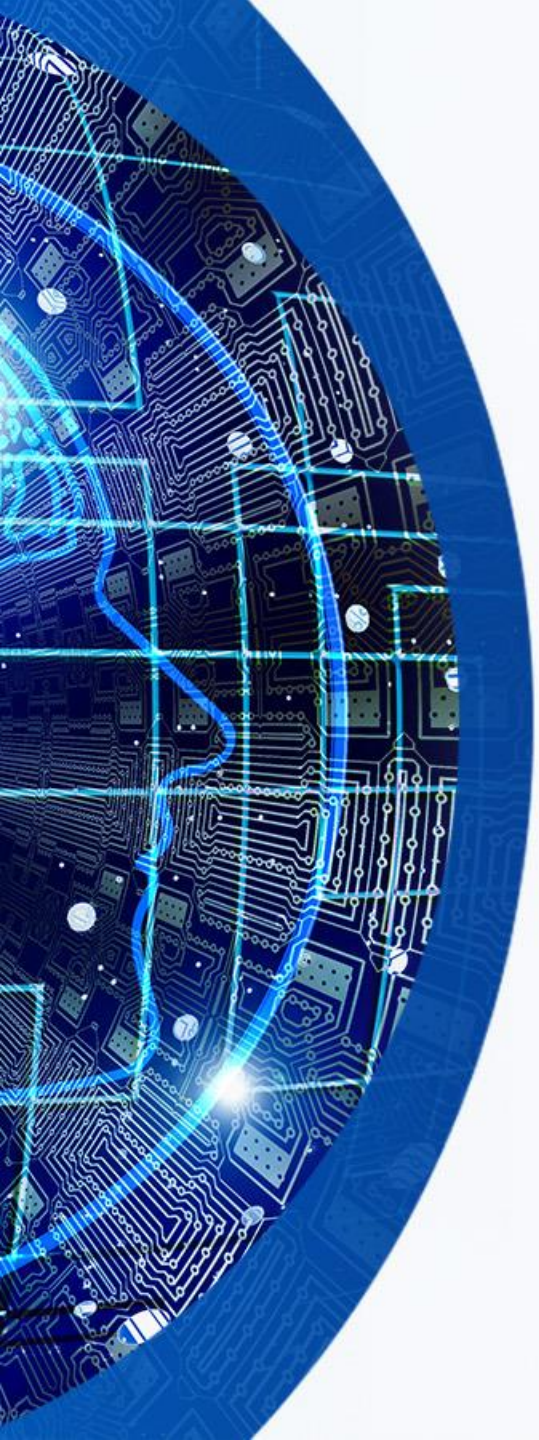
Indexing

- Term weighting creates a distinction among terms and increases indexing flexibility.
- Thus we need to assign **higher weight** to more "**important**" features and **lower** weight to **marginal** ones.
- To weight appropriately each indexing unit, we may consider three components, namely,
  - the term frequency ***tf***,
  - the document frequency ***df***,
  - and the document length.



Indexing

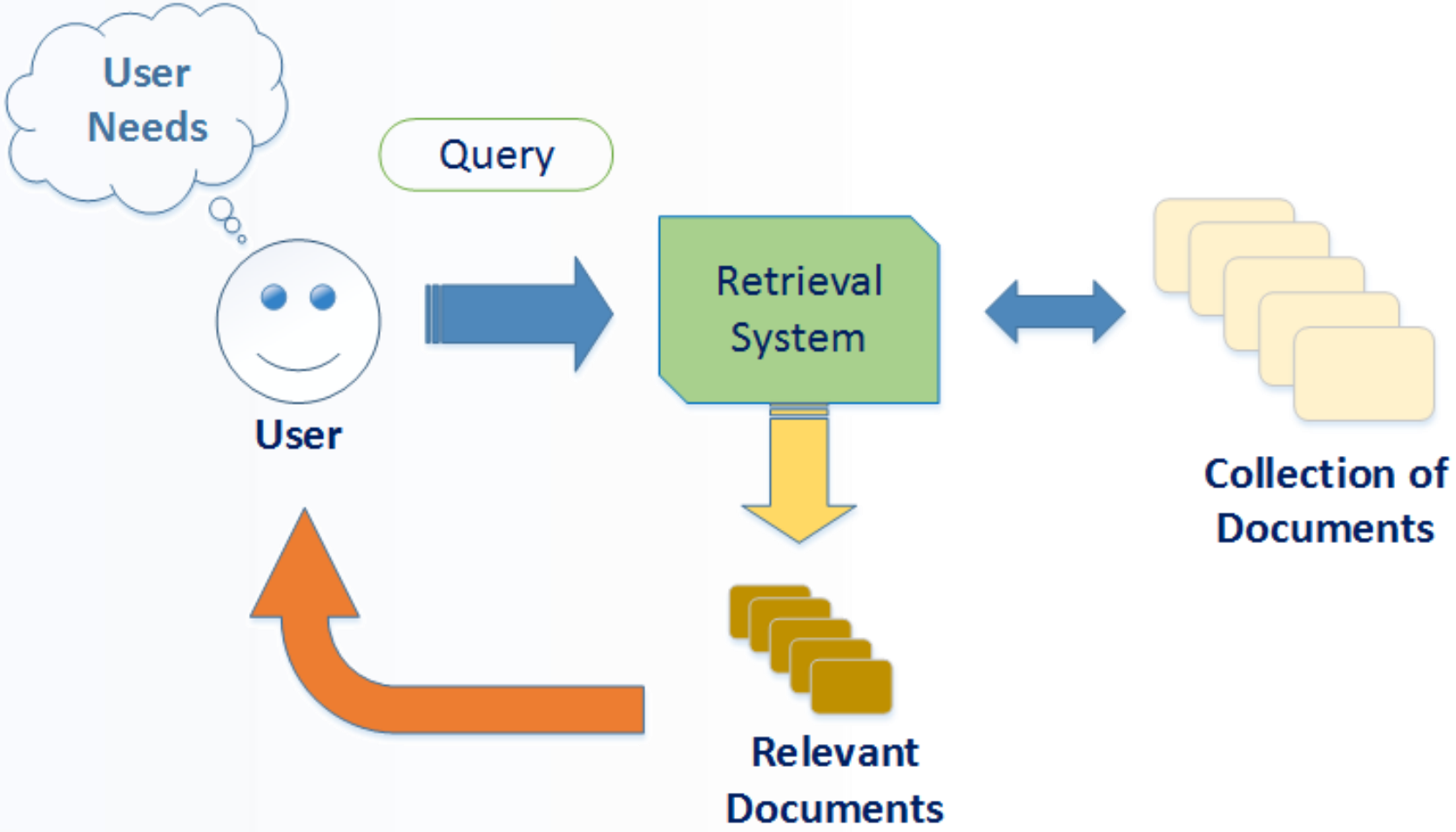
- **TF**; First, one can assume that an indexing unit appearing **more often** in a document must have a higher importance in describing its semantic content,
- **DF**; As a second weighting component, one may consider that those terms occurring very **frequently** in the collection do not help us discriminate between relevant and nonrelevant documents.
- **Doc. Length**; Lastly, one usually considers that the presence of a term in a **shorter** document provides stronger evidence than it does in a longer document.

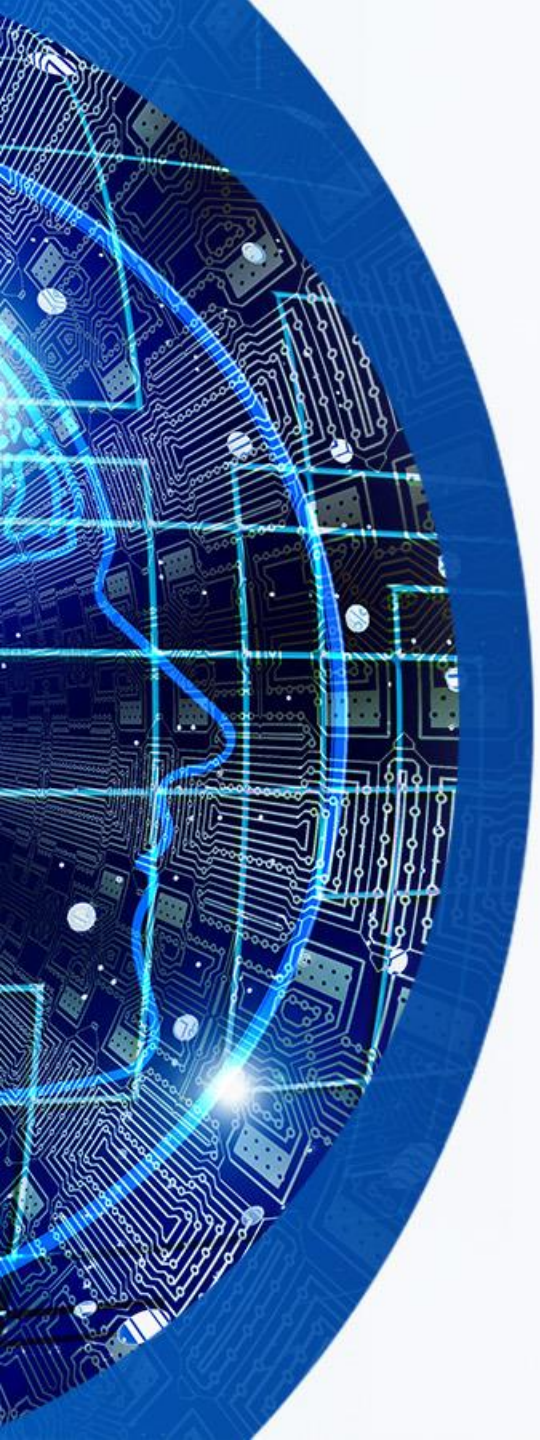


Query Expansion

- To provide better matching between **user information needs** and **documents**, various **query expansion** techniques have been suggested.
- The general principle is to **expand** the query **using words or phrases** having meanings **similar** or **related** to those appearing in the original query, either by using information from a thesaurus or by **deriving** this information from the collection.

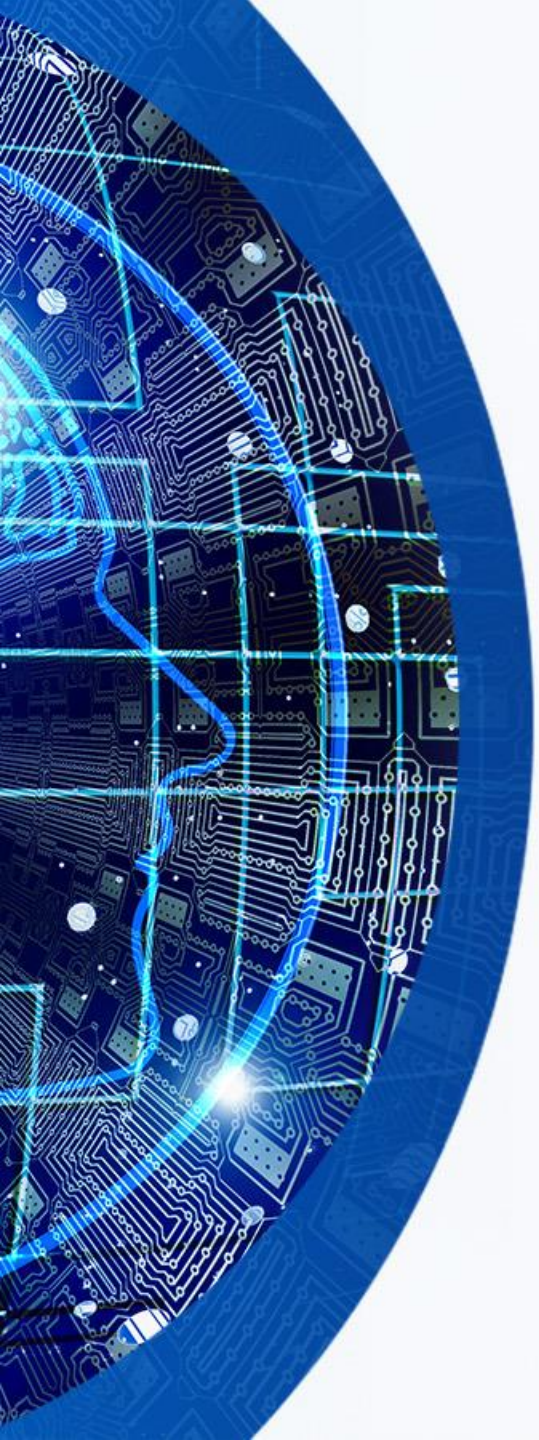
NLP & IR


User Needs
Query
User
Retrieval System
Collection of Documents
Relevant Documents



NLP & IR

- The basic building blocks of most natural languages are words.
- However, the term "**words**" is ambiguous and we must be more precise in order to distinguish between the **surface form** (e.g., "horses") corresponding to tokens, and word type or lemma (entry in the dictionary, such as "horse" in our case).
- Moreover, the specific meaning of the term "horse" is not always an animal with four legs as in the term "**horse-fly**" or in expressions as "**Trojan horse**," "**light horse**," "**to work like a horse**," "**from the horse's mouth**," or "**horse about**".

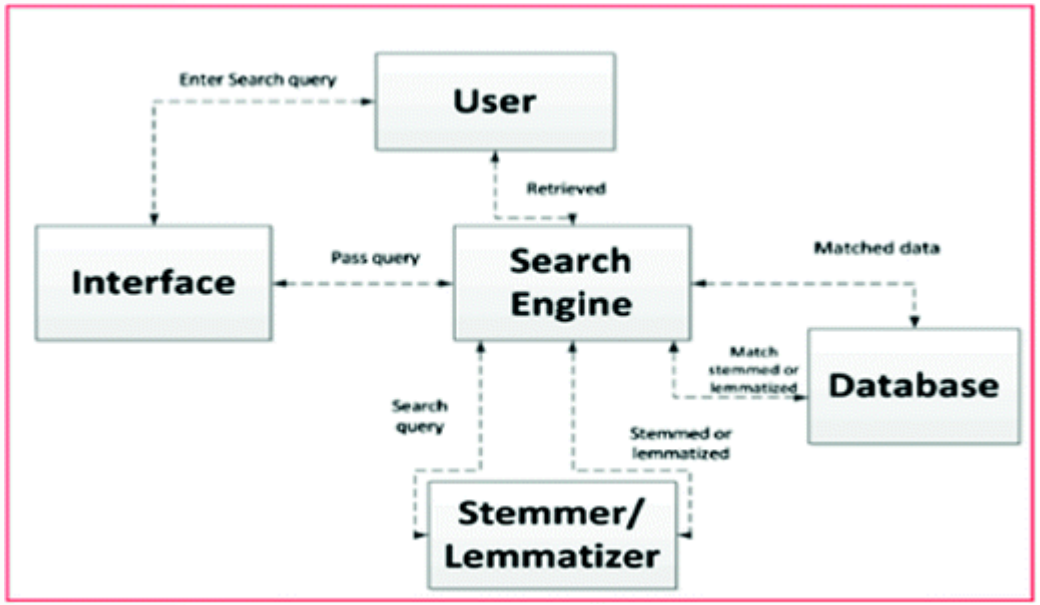


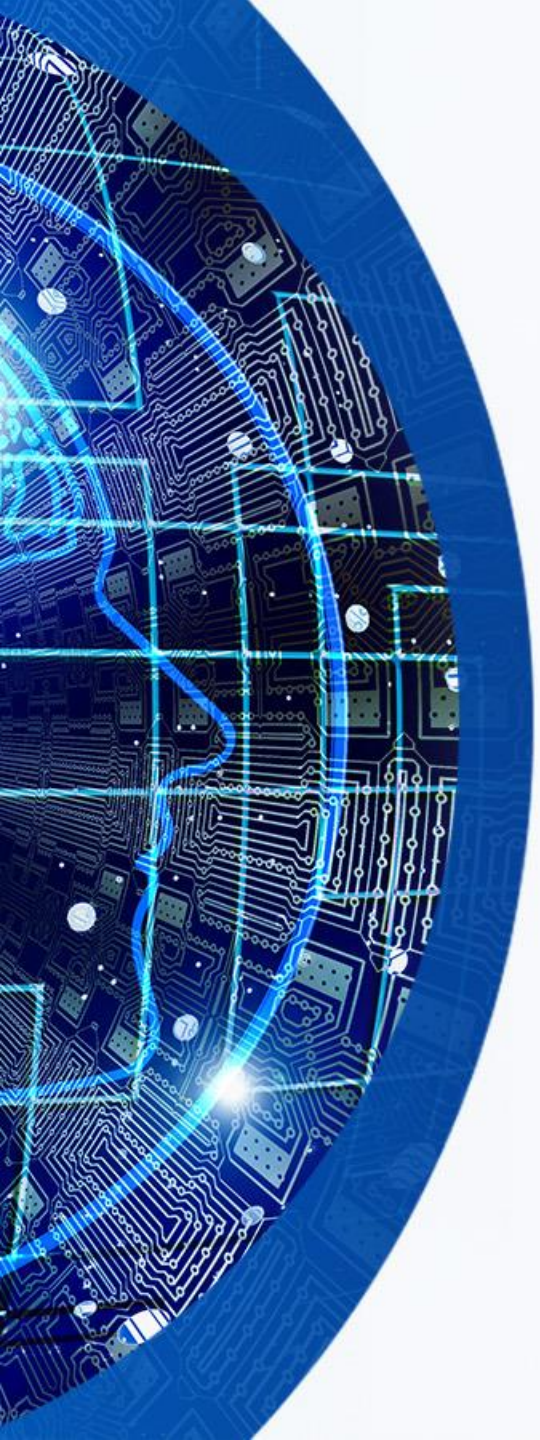
NLP & IR

- In the design of effective IR systems, the **morphological** component plays an important role.
- For most search systems, and also for most human languages, **the words** form the basic units to build the phrases, expressions, and sentences used to transmit a precise meaning.
- An appropriate **processing of these entities** is therefore important for enhancing **retrieval performance** by promoting pertinent **word sense matching.**
- In addition, the way **words combine** and the meanings words convey are crucial for understanding the content of a document and for representing this content accurately.

NLP & IR

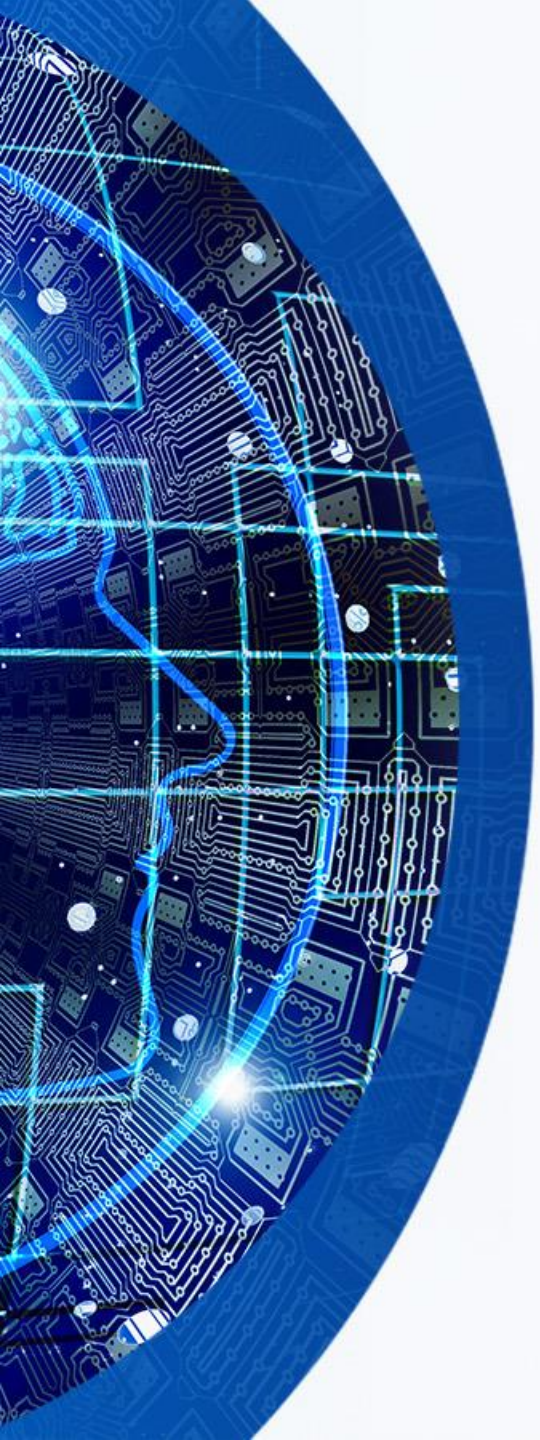
- In the next slides, we will review some of the major **relations** between IR and NLP.
- We will do so by examining the traditional layers of Natural Language Text Processing (**morphology**, **syntax**, **semantics**) and see the **role** they play or can play in IR.
- Finally, we will briefly mention various **applications** that have direct connections with both IR and NLP





Morphology,

- The goal of the **morphological** step in IR is to conflate morphological **variants into the same form**.
- In some cases, only an **inflectional** (ed, es, ing…..)
- analysis is performed, so as to get to a lemmatized version of the original text.
- This step can be followed by a **derivational** analysis, usually relying on **suffixes** only, as **prefixes** tend to radically modify the meaning of a word (indexing the two forms "**decompose**" and "**recompose**" as the same token does not make sense from an IR point of view).

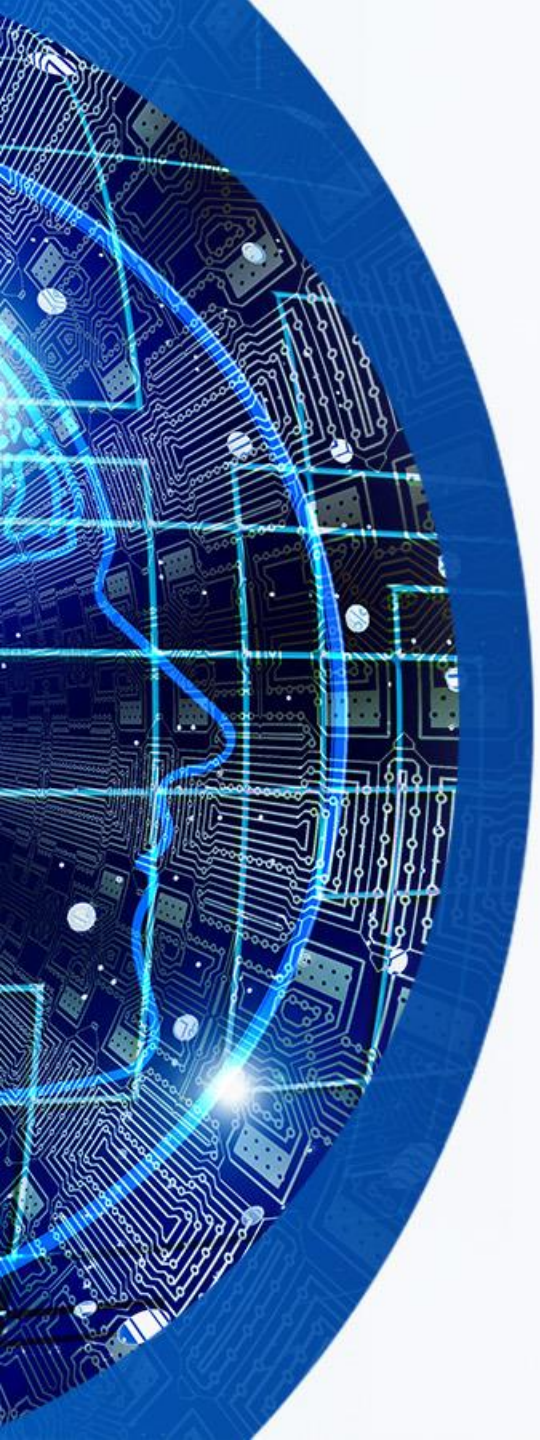


Morphology,

- When defining a **stemming** algorithm, a first approach will only remove **inflectional** suffixes
- For English, such a procedure conflates singular and plural word forms (“car” and “cars”) as well as removing the past participle ending “-ed” and the gerund or present participle ending “-ing” (“eating” and “eat”).
- Stemming **schemes** that remove only morphological inflections are termed as “**light**” suffix-stripping algorithms, while more **sophisticated** approaches have also been proposed to remove derivational suffixes (e.g., “-ment,” “-ably,” “-ship” in the English language).

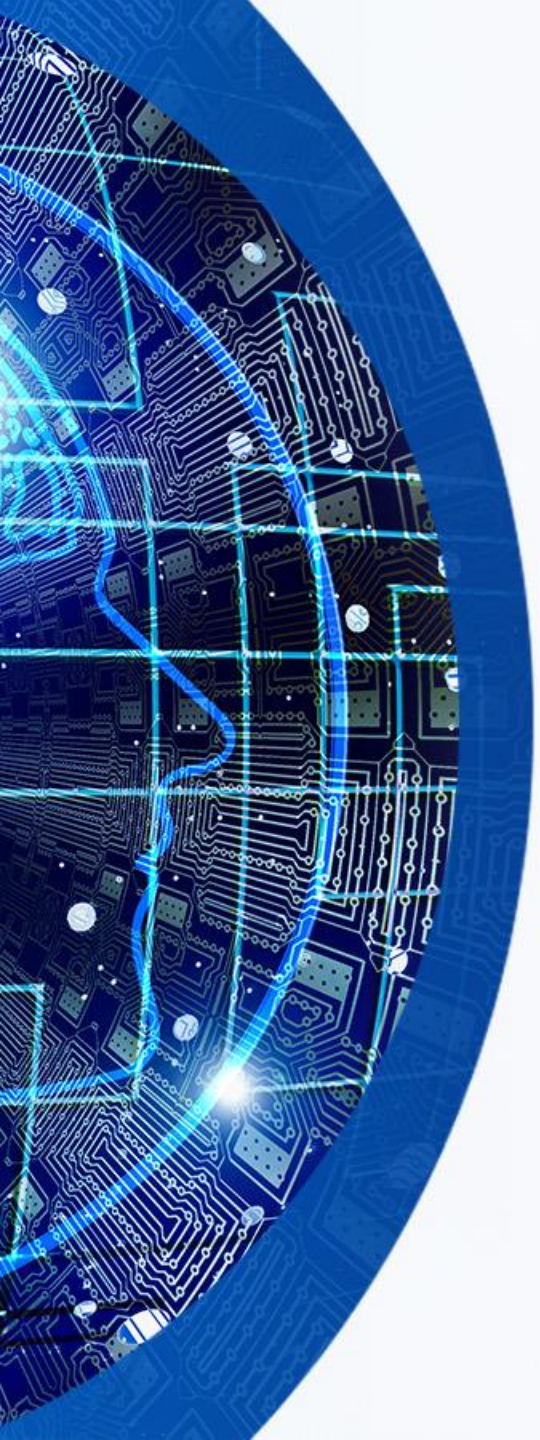
Morphology,

- For example, **Lovins** (1968) is based on a list of over **260 suffixes,** while Porter's algorithm (**Porter**, 1980) looks for about **60 suffixes**.
- In such cases, suffix removal is also controlled through the adjunct of :
  - **quantitative** restrictions (e.g., "-ing" would be removed if the resulting stem had more than three letters as in "**running**," but not in "**king**") or
  - **qualitative** restrictions (e.g., "-ize" would be removed if the resulting stem did not end with "e" as in "seize" bu not in 'criticize').



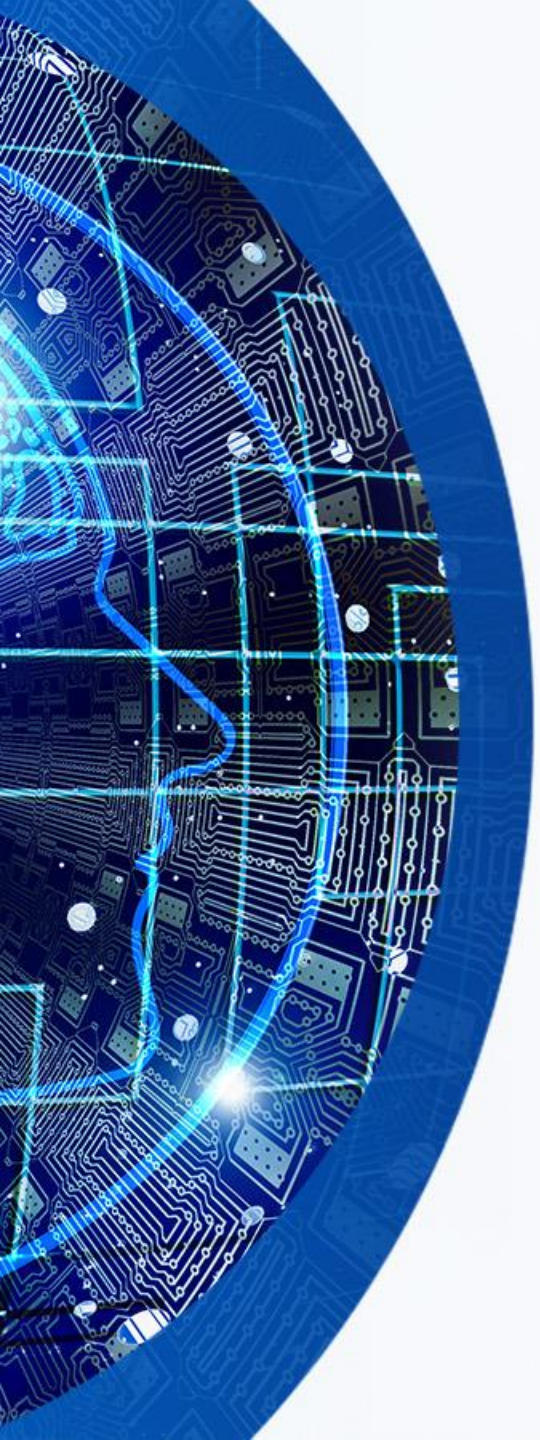
Morphology,

- Stemming schemes are usually designed to work with **general text** in any given language.
- Certain stemming procedures may also be **especially designed** for a specific domain (e.g., in medicine) or a given document collection.
- stemming procedures **ignore** word meanings, and thus tend to make **errors**.
- Such errors may be due to over-stemming (e.g., “general” becomes “**gener**,” and “organization” is reduced to “organ”) or under-stemming (e.g., with Porter’s stemmer, the words “create” and “creation” do not conflate to the same root)



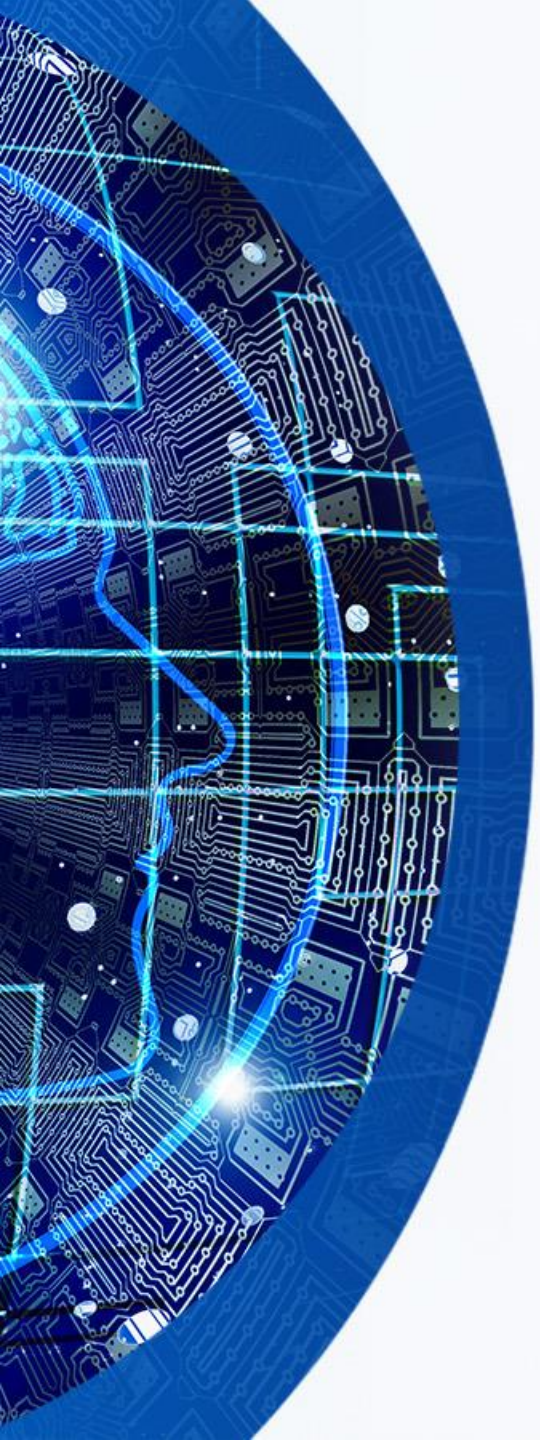
Morphology,

- The development of a morphological analyzer (be it a stemmer or a more refined tool) **depends** largely on the **language** considered.
- The **English** inflectional morphology is relatively simple.
  - The **plural** form is usually denoted by adding an ’-s’ (with some exceptions like “foot” and “feet”).
  - The **feminine** form is built using some suffixes (e.g., “actor” and “actress”) and we do not have to mark the agreement between noun and adjective (“tall man,” “tall women”).
  - To build new words (derivational construction), we may add **prefixes** (“pay,” “prepay”) and/or suffixes (“bank,” “banking,” “banker,” “bankless,” “bankrupt”).



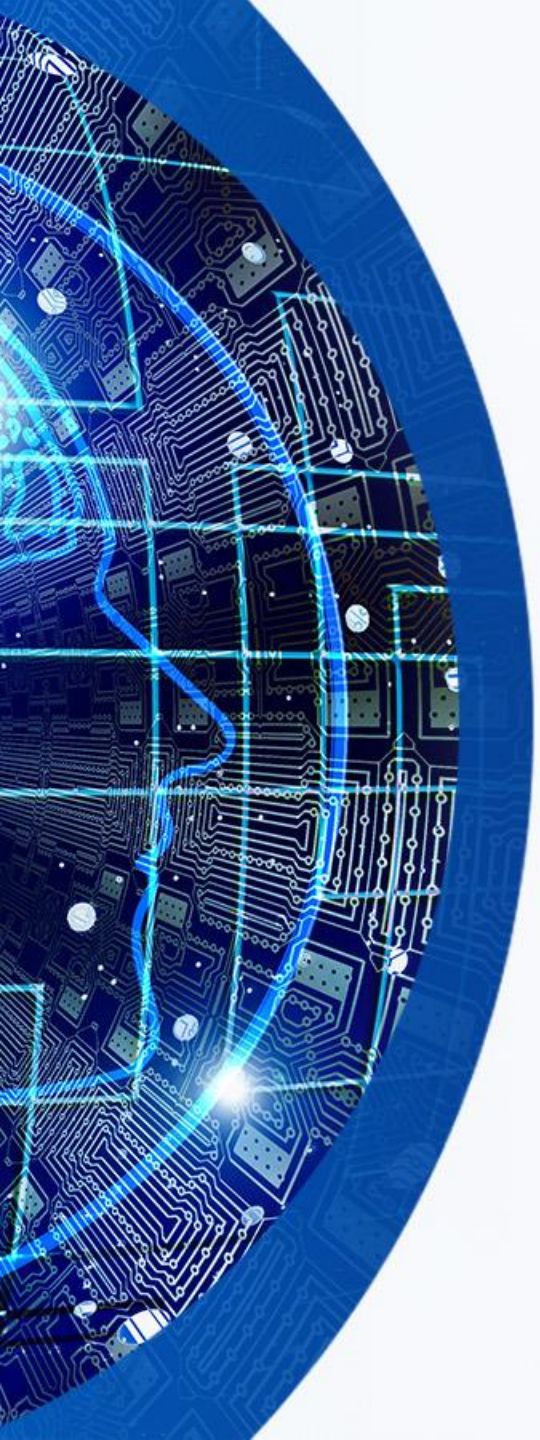
Morphology,

- For other **European** languages, the morphological construction can be relatively similar.
- In the **French** language,
  - the **plural** is denoted as in English (“face,” “faces”),
  - while the **feminine** form can simply be denoted by a final ‘**-e**’ (“employé,” “employée”) or by a suffix (“acteur,” “actrice”).
  - As in English, various suffixes are available to form new words



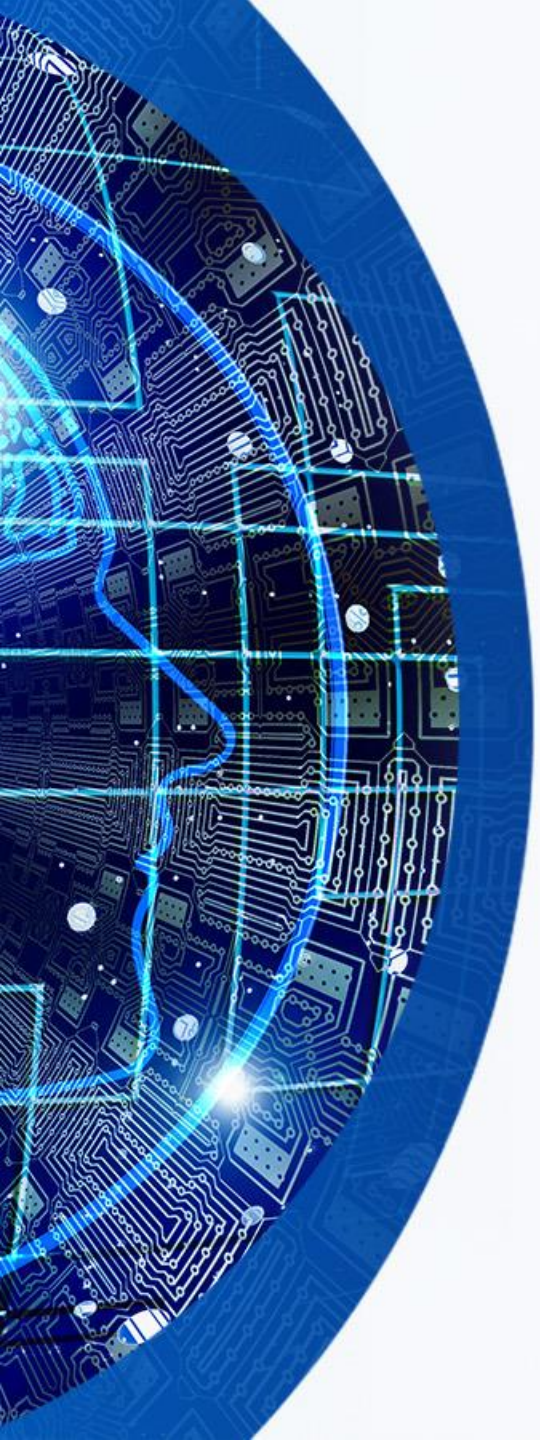
Morphology,

- **Compound** constructions (**concatenation** of two or more **lexemes** to form another word, e.g., **handgun**, **worldwide**) also appear in other European languages.
  - In Italian, the **plural** form may alter letters within a word, for example, "**capoufficio**" (chief secretary) becomes "**capiufficio**" in its plural form.
  - Yet, in other constructions, the stem "capo" is left unchanged (e.g., "capogiro" gives "capogiri" (dizziness) in its plural form)
- Several tools are available for identifying morphological variants in different languages.
- They are either based on online **dictionaries** and **standard** morphological analysis or on **stemming** procedures **dedicated** to different languages



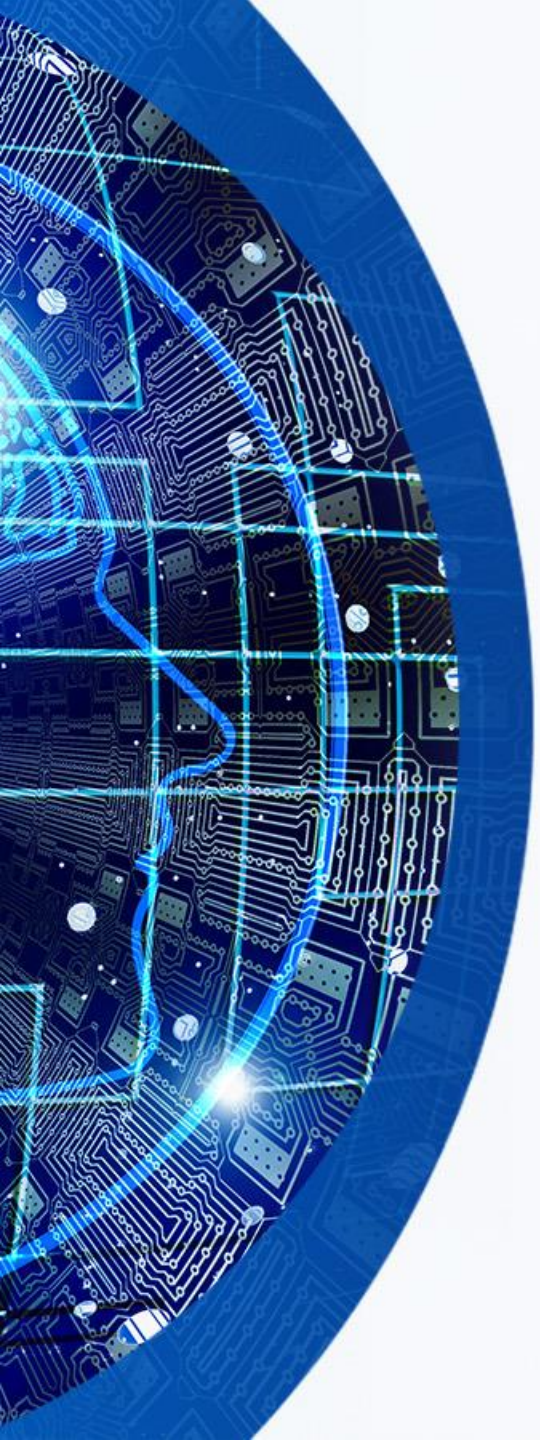
Orthographic Variation and Spelling errors,

- The **standardization** of spelling was mainly the fruit of the nineteenth century.
- Working with documents **written** previously, one encounters different spellings for a given term or proper name
  - e.g., Shakespeare's name appears as "Shakper," "Shakspe," "Shaksper," "Shakspere," or "Shakspeare" in his own works.
- The spelling problem can be **domain-specific**.
- In the **biomedical** literature, it is known that several **orthographic** variants can be found to represent a given **name**, generally introduced for a variety of reasons.



Orthographic Variation and Spelling errors,

- **Firstly**, there are of course typographic errors and misspellings (performance errors as in “retreival” and “retrieval” or competence errors as in “exam,” “ecsam,”
- **Secondly**, punctuation and tokenization may produce variants, mainly due to the lack of a naming convention (e.g., “Nurr77,” “Nurr-77,” or “Nurr 77”).
- **Thirdly**, regional variations also introduce variants (e.g., the difference between British and American English for “colour” and “color” or “grey” or “gray”).
- **Fourthly**, the transliteration of foreign names produces some differences (e.g., “Crohn” and “Krohn” or “Creutzfeld-Jakob” and “Creutzfeldt-Jacob”).

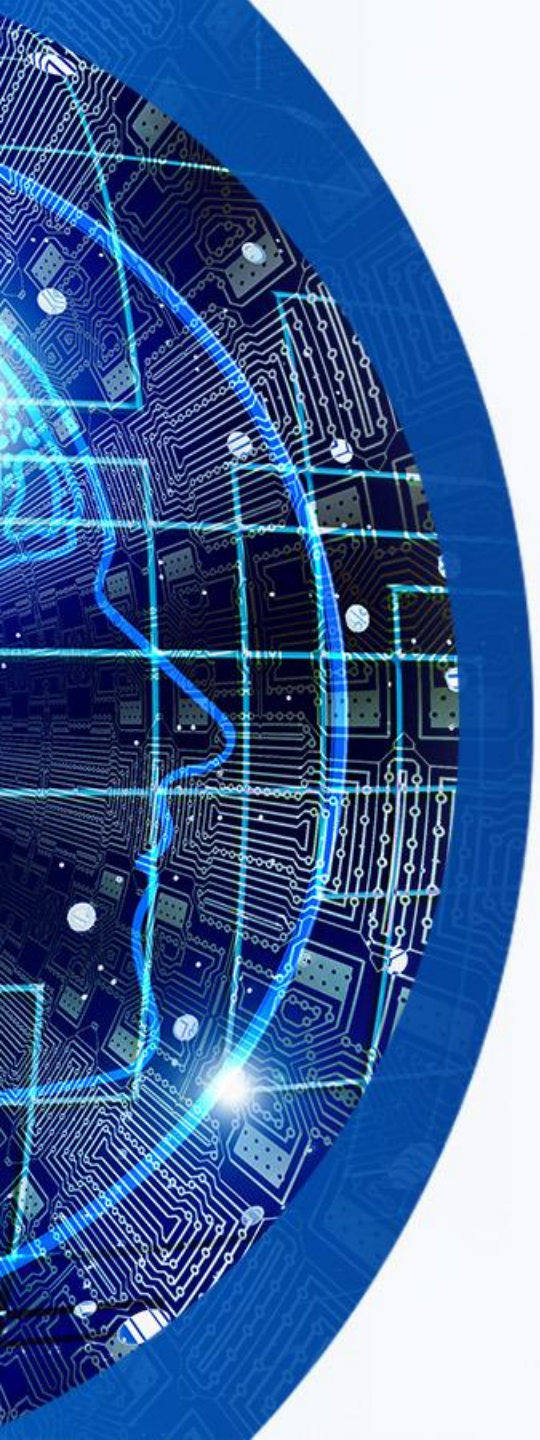


Orthographic Variation and Spelling errors,

- The **standard strategy** to reduce the negative impact caused by spelling errors or orthographic variation is to **relate** similar (however not identical) forms in one way or another.
- **Two** main **strategies** can be used here:

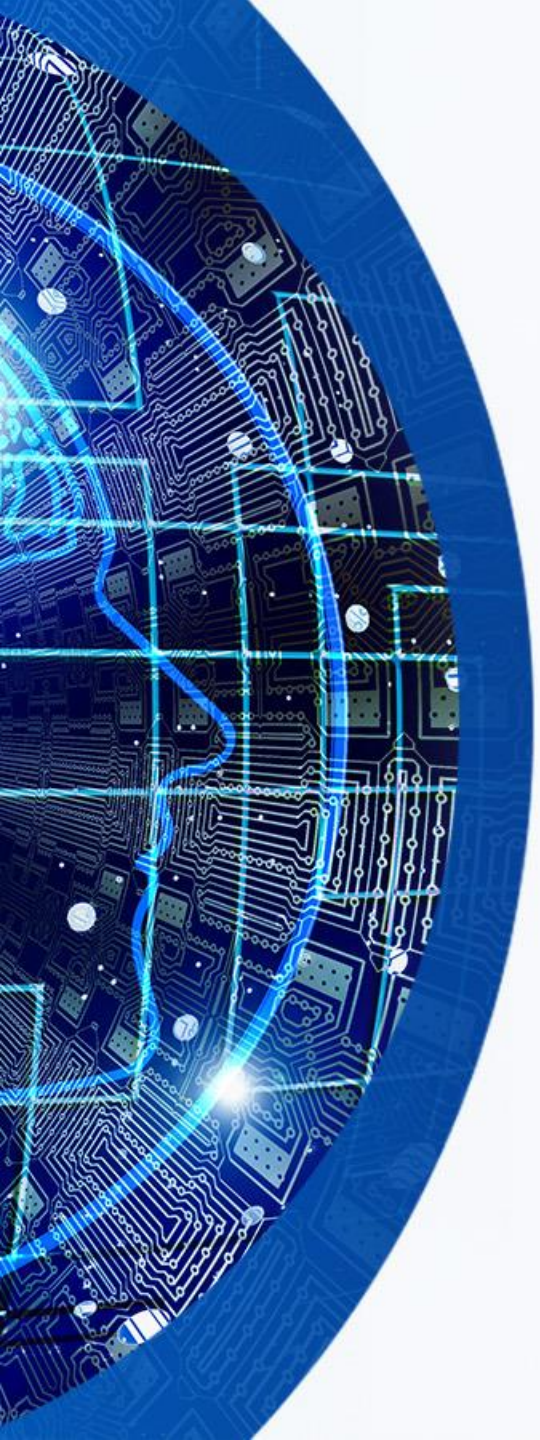
  (a) compute an **edit-distance** between different forms (e.g., present in a dynamic dictionary) and normalize the variants with a particular form,

  (b) adopt an **n-gram** indexing strategy (n is typically set to five across a range of studies).



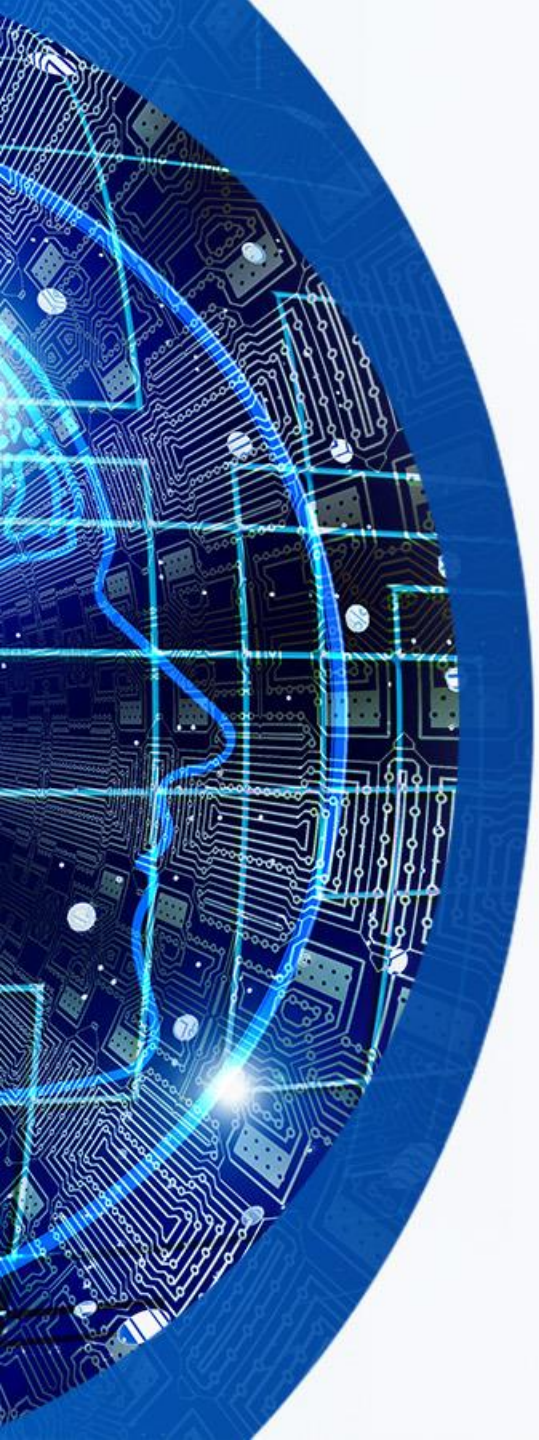
Orthographic Variation and Spelling errors,

- The **n-gram** method has the advantage of being fast (filters have to be used to avoid **comparing unrelated** forms in the first approach).
- Moreover, this method does not require any prior linguistic knowledge and is robust to typographical errors, both in the submitted queries and documents retrieved.
- For instance, the term “alzheimer” would be decomposed, with an overlapping five-gram approach, as: “alzhe,” “lzhei,” “zheim,” “heime,” and “eimer.”



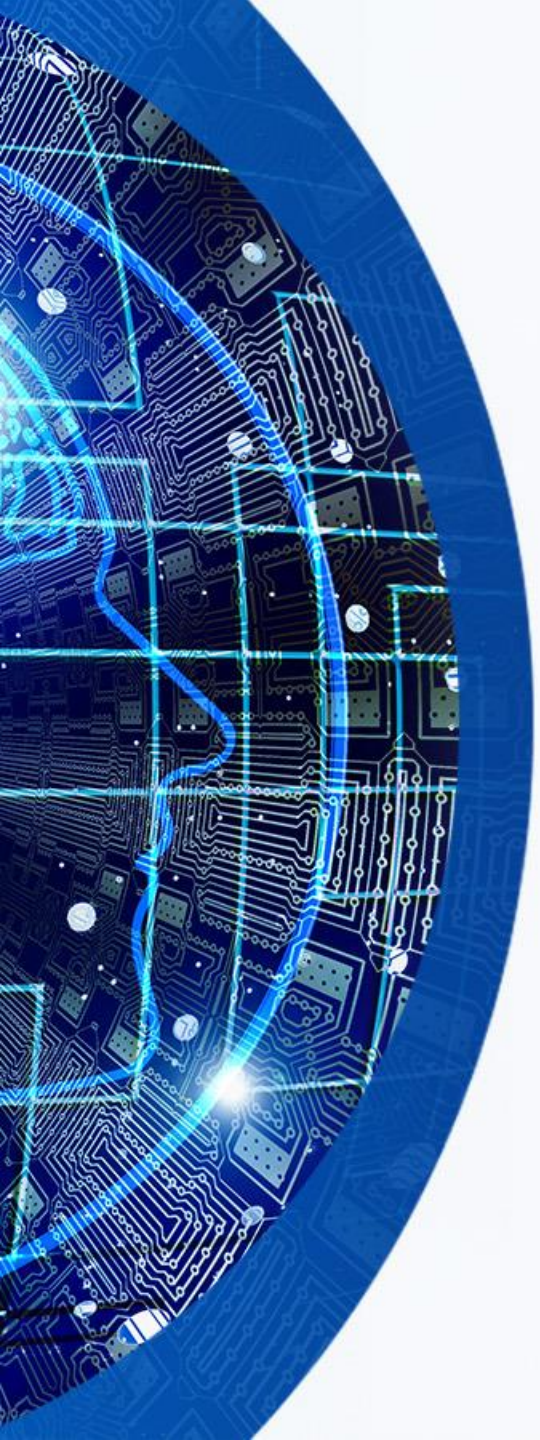
Syntax

- Most IR systems **index** documents on the basis of the **simple** words they contain.
- However, there have been many attempts to make use of more **complex** index terms, comprising several terms, in order to get a more **precise** description of the content of documents.
- One of the first attempts, referred to as adjacent pairs, considered complex terms made up of **two adjacent** content words.



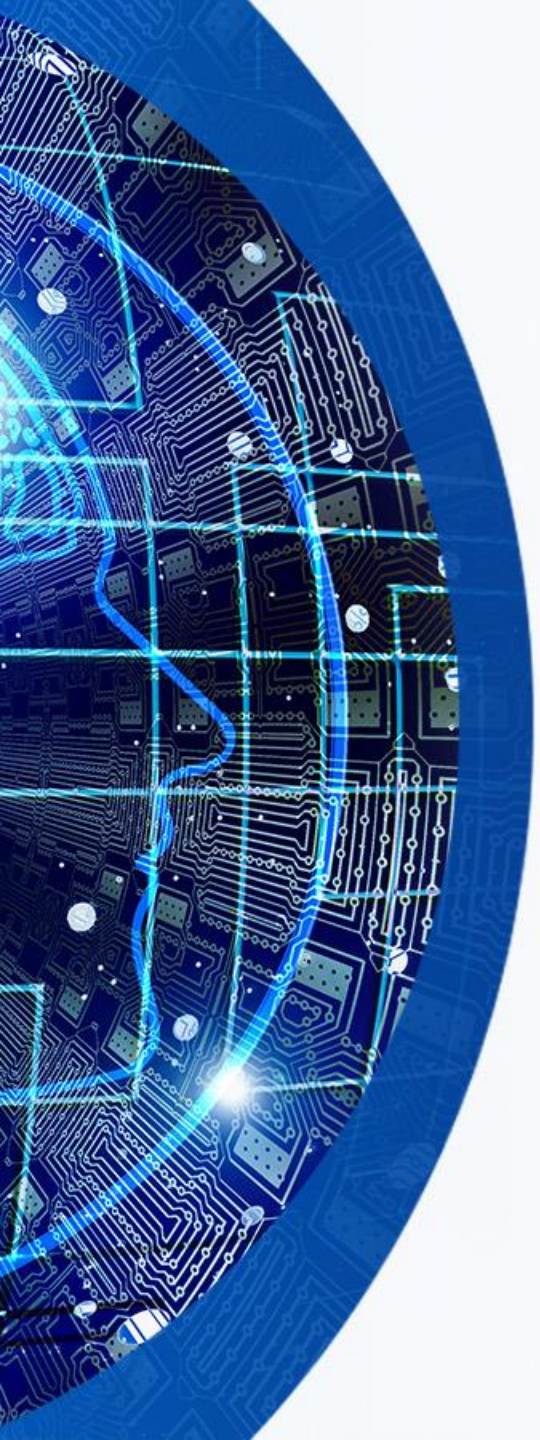
Syntax

- Even though simple, this approach is appropriate e.g., for the English language, as it allows one to capture terminological elements consisting of **Adjective Noun** or **Noun Noun** sequences (and their combination).
- For other languages, such as the ones based on a composition of Romance type, i.e., in which compounds are formed through **prepositions**, simple regular expressions can be used over **part-of-speech tags** to identify terminological elements and add them to the index terms.
- More generally, a **syntactic analyzer** can be used to identify long-distance dependencies between words, so as to improve the indexing of documents.



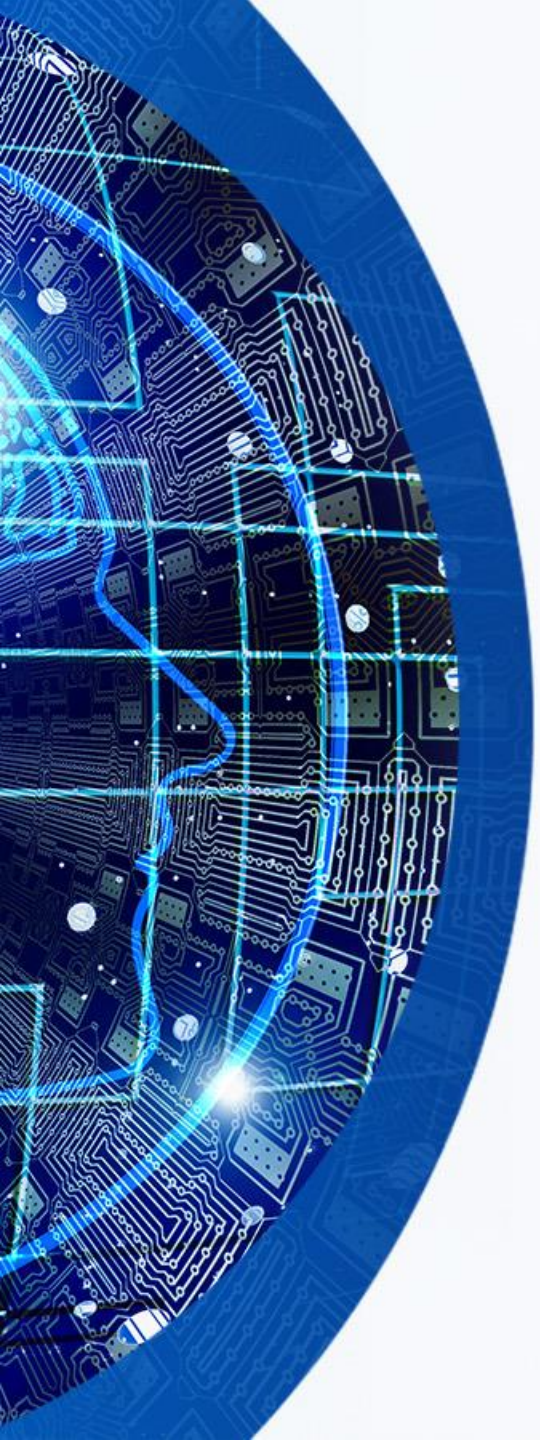
Syntax

- However, the lack of **large-coverage** grammars, and the difficulty of obtaining **unambiguous** and correct parses for many sentences effectively put a stop to this type of research at that time
- Advances in shallow (or light) parsing in the mid-1990s led to a resurgence of this type of work, as such parsers were partly defined to be general and robust.
- for example, to use a **shallow parser** to identify relations within and between complex noun and verb phrases.
- **Pairs of terms** thus **extracted** (as Subject Verb pairs or Adjective Noun pairs) are **then added to simple words** to provide a richer index of documents.



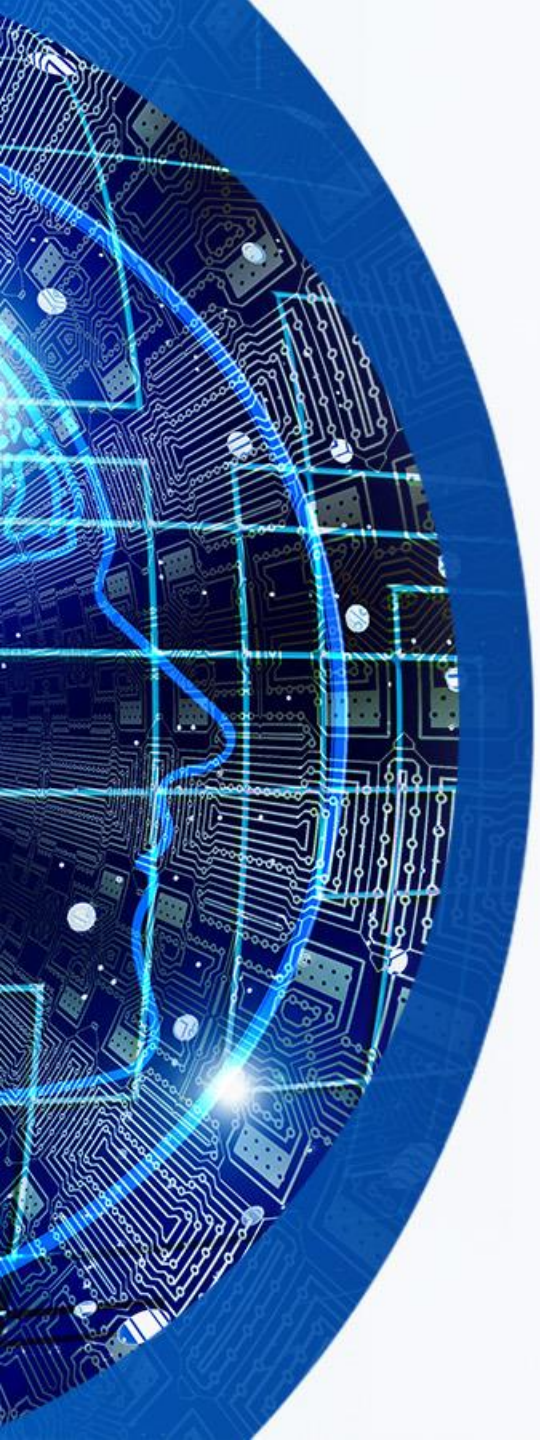
Syntax

- Similar experiments conducted on French, proposed an extension of the language modeling approach to IR that can take into account the **syntactic** dependencies provided by a probabilistic parser deployed on the queried collection.
- **Complex terms** acquired through syntactic analysis can be seen as **additional information** that can help **refine query results** and thus lead to an increase in the precision at the top of the list of retrieved documents.
- However, in many cases, having constituents of a compound taken independently of each other or as a whole **does not change** the retrieval results substantially.



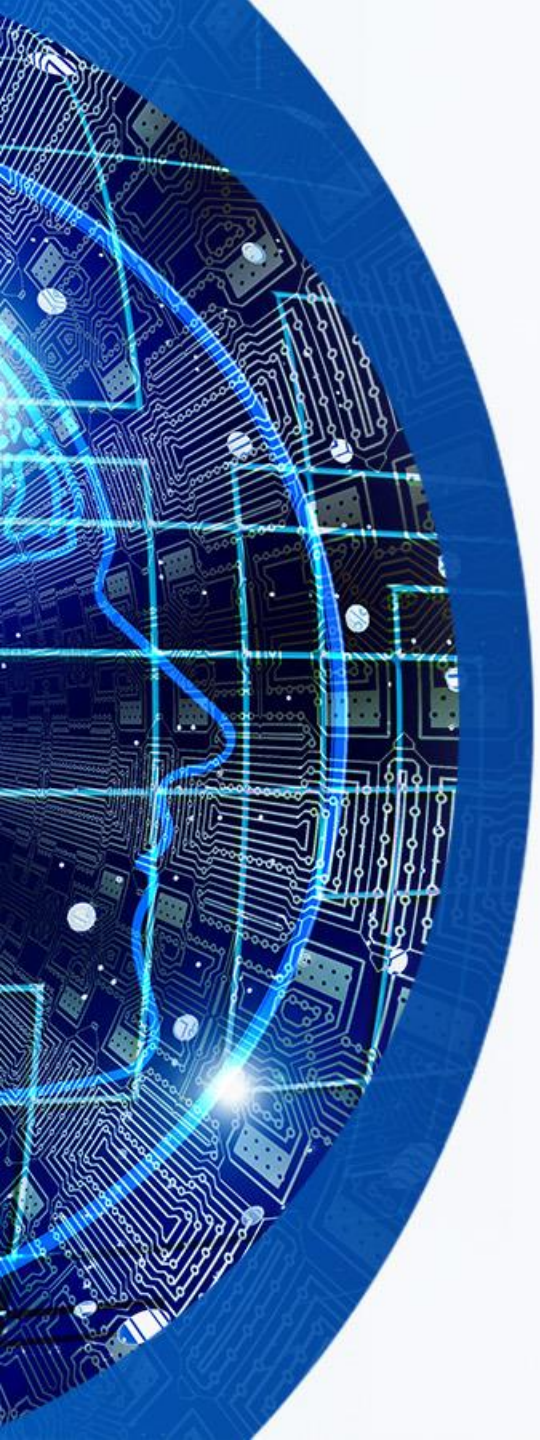
Syntax

- For example, a document indexed with the two terms **information** and **retrieval** is very likely to deal with **information retrieval**, so that the addition of this last compound does not really add any new information: it will be indexed by information retrieval after a syntactic analysis phase, and the ranking will not change
- In essence, syntactic analysis provides additional indexing dimensions, **but does not address the problem** of the vocabulary mismatch between queries and documents.
- A direct way to address this problem is to resort to a **semantic** analysis in order to replace index terms with concepts and perform the matching between queries and documents at a more abstract level



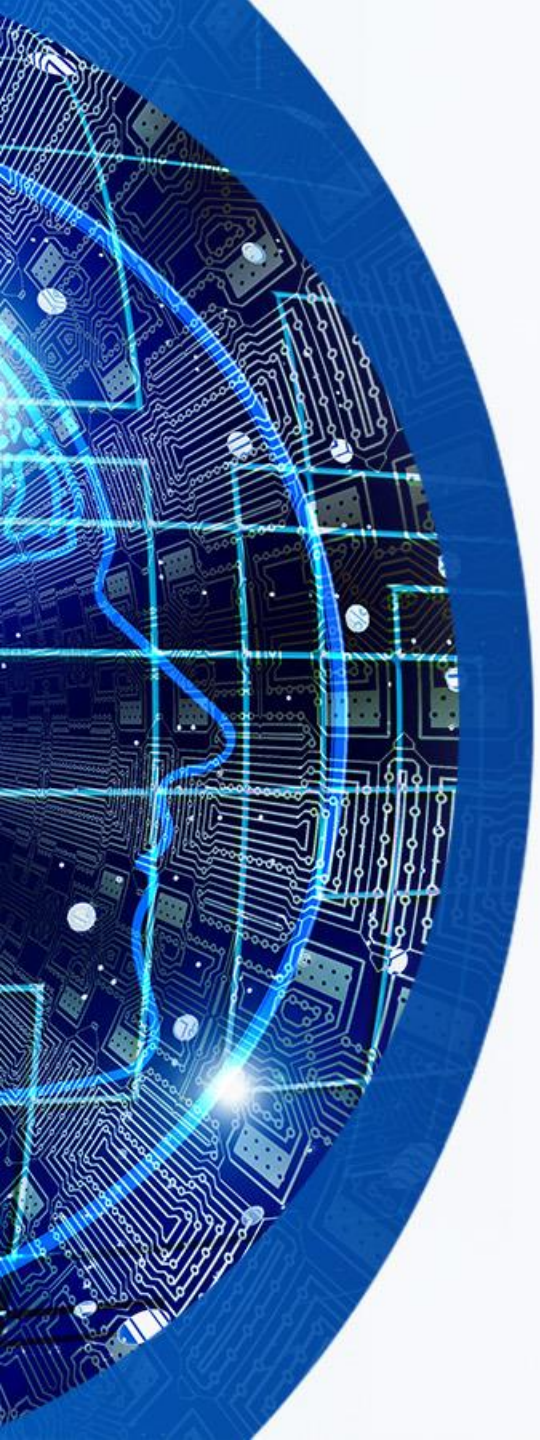
Semantics

- In the absence of **robust systems** providing a complete semantic analysis of sentences, most work in IR and semantics have **focused on lexical semantics**, and the possibility to **replace standard index** terms with **concepts or word senses.**
- While some studies have shown that **word sense disambiguation procedures** can be beneficial in some cases, the majority have tried to rely on existing semantic resources in order to index both documents and queries at the concept level.
- Such works have used existing **semantic resources**, either **generic** ones, or **specialized** ones for specific collections



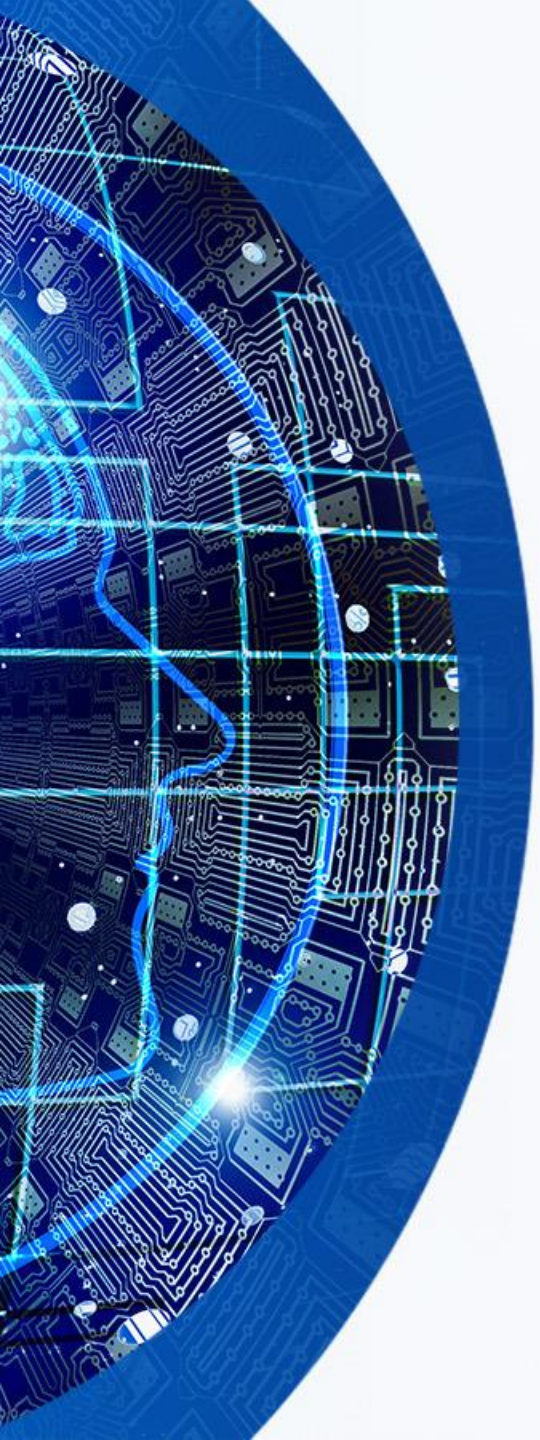
Semantics

- **Lexical semantics** is concerned with understanding how individual words and phrases contribute to the overall meaning of a sentence or discourse
- **word sense disambiguation procedures,** consider the two sentences. “The bank will not be accepting cash on Saturdays. ” “The river overflowed the bank.” The word bank in the first sentence refers to the commercial (finance) banks, while in second sentence, it refers to the river bank.



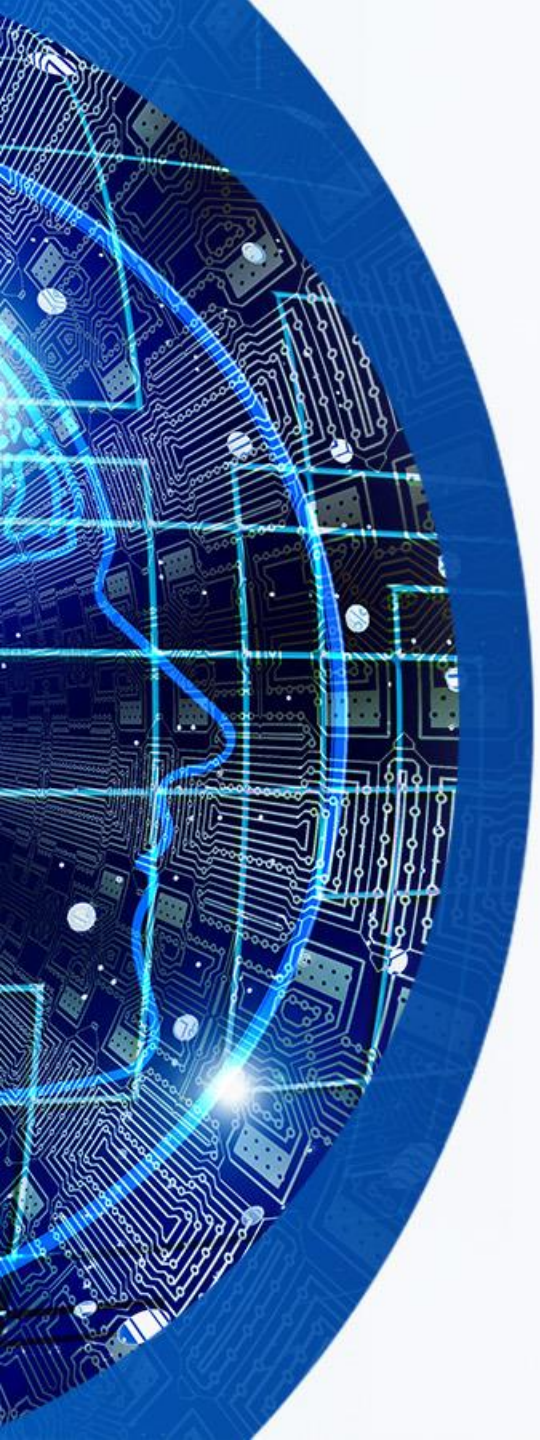
Semantics

- The **usefulness** of concept indexing in **specific domains** has been shown in several studies, mainly in the **medical domain** for which large-coverage thesauri and semantic networks exist.
- For example, the **best-performing** systems used **thesaurus and databases** to select terms from medical publications.
- Terms in documents and **queries were expanded with their variants found in some resources** to achieve better indexing.
- They found that this strategy led to a **significant improvement over standard indexing methods.**



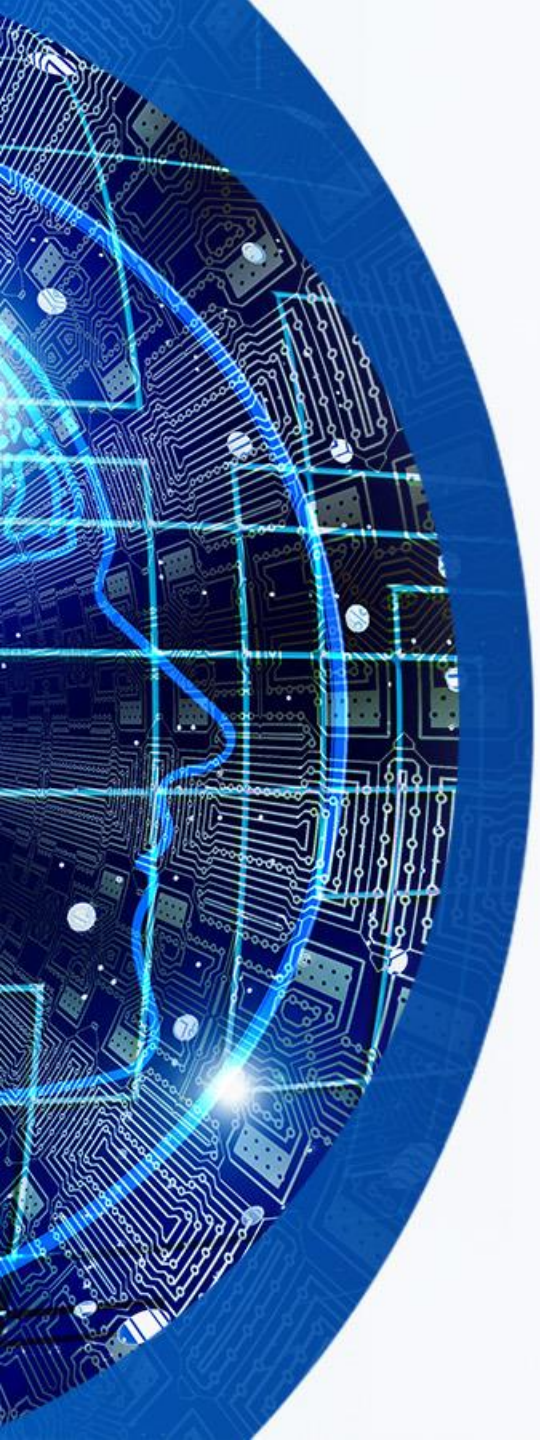
Semantics

- Other researchers have tried to go beyond the use of concepts by exploiting relations between them, they evaluates the usefulness of UMLS concepts and semantic relations in medical IR.
- Unified Medical Language System
- They first extract concepts and relations from documents and queries. To select relations in a sentence, they rely on two assumptions:
- (1) interesting relations occur between interesting concepts;
- (2) relations are expressed by typical lexical markers such as verbs.



Conclusion

- the IR field is an extensive applied NLP domain that is able to cope successfully with search and retrieval in the huge volume of information stored on the Web—users are, on average, able to find what they are looking for.
- This success may however hide other challenging problems.
- Commercial search engines want to know more about the users and their needs to have a better knowledge of the real question behind the submitted query.
- This information can play an important role in commercial advertising.



Conclusion

- From this perspective, the search system will try to complement its answer page with advertising banners that are appropriate to both the user query and preferences.
- Finally, the relationship between the IR and NLP domains tends to be strong on one hand, and multifaceted on the other.
- As presented in this chapter, morphology, spelling error correction, syntax, and semantics are important aspects for general or domain-specific IR systems.